

A Survey on Spatio-Temporal Prediction: From Transformers to Foundation Models

YINGCHI MAO, Hohai University, Nanjing, China

HONGLIANG ZHOU, Hohai University, Nanjing, China

LING CHEN, Hohai University, Nanjing, China

RONGZHI QI, Hohai University, Nanjing, China

ZHENDE SUN, Nanjing University of Posts and Telecommunications, Nanjing, China

YI RONG, Hohai University, Nanjing, China

XIAOMING HE, Nanjing University of Posts and Telecommunications, Nanjing, China

MINGKAI CHEN, Nanjing University of Posts and Telecommunications, Nanjing, China

SHAHID MUMTAZ, Nottingham Trent University, Nottingham, United Kingdom of Great Britain and Northern Ireland

VALERIO FRASCOLLA, Intel Deutschland GmbH, Neubiberg, Germany

MOHSEN GUIZANI, Mohamed bin Zayed University of Artificial Intelligence, Masdar City, United Arab Emirates

JOEL RODRIGUES, Federal University of Piauí, Teresina, Brazil

Spatio-Temporal (ST) data is pervasive on the various aspects in our daily lives. By mining the ST information from the data, we are able to predict trends in numerous domains. The Transformer and one of its more recent enhancements, foundation models, have achieved a remarkable success in such ST prediction. In this article, we first survey the state of the art of Transformers-related work, then introduce the network architecture of the Transformer and summarize the improvements to adapt to the ST prediction Transformer and foundation models, including module enhancement and adjustment. Subsequently, we categorize the

Yingchi Mao, Hongliang Zhou, and Rongzhi Qi contributed equally to this research.

This work was supported in part by the Key Research and Development Program of China under Grant 2022YFC3005401; the Technology Talent and Platform Program of Yunnan Province under Grant 202405AK340002; in part by the Technology Project of Huaneng Group under Grant HNKJ24-H167; in part by the High Performance Computing Platform, Hohai University; in part by Brazilian National Council for Scientific and Technological Development - CNPq, via Grant No. 306607/2023-9.

Authors' Contact Information: Yingchi Mao, Hohai University, Nanjing, Jiangsu, China; e-mail: yingchimao@hhu.edu.cn; Hongliang Zhou, Hohai University, Nanjing, Jiangsu, China; e-mail: zhouhongliang@hhu.edu.cn; Ling Chen, Hohai University, Nanjing, China; e-mail: 221307050018@hhu.edu.cn; Rongzhi Qi, Hohai University, Nanjing, Jiangsu, China; e-mail: rzqi@hhu.edu.cn; Zhende Sun, Nanjing University of Posts and Telecommunications, Nanjing, Jiangsu, China; e-mail: 1023010439@njupt.edu.cn; Yi Rong, Hohai University, Nanjing, Jiangsu, China; e-mail: rongyi1220@163.com; Xiaoming He, Nanjing University of Posts and Telecommunications, Nanjing, Jiangsu, China; e-mail: hexiaoming@njupt.edu.cn; Mingkai Chen (corresponding author), Nanjing University of Posts and Telecommunications, Nanjing, China; e-mail: mkchen@njupt.edu.cn; Shahid Mumtaz, Nottingham Trent University, Nottingham, United Kingdom of Great Britain and Northern Ireland; e-mail: dr.shahid.mumtaz@ieee.org; Valerio Frasca, Intel Deutschland GmbH, Neubiberg, Bayern, Germany; e-mail: valerio.frascola@intel.com; Mohsen Guizani, Mohamed bin Zayed University of Artificial Intelligence, Masdar City, Abu Dhabi, United Arab Emirates; e-mail: mguizani@ieee.org; Joel Rodrigues, Federal University of Piauí, Teresina, PI, Brazil; e-mail: joeljr@ieee.org.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM 0360-0300/2025/10-ART85

<https://doi.org/10.1145/3766546>

ST Transformer and foundation models in selected applications in some relevant domains, mainly urban transportation, climate monitoring, and motion prediction. Next, we propose an evaluation method in the ST prediction with Transformers and foundation models, list the most relevant open-source datasets, evaluation metrics and performance analysis. Finally, we discuss some future directions on the task of ST prediction with Transformer and foundation models. Relevant papers and open-source resources have been collated and are continuously updated at: <https://github.com/cyhforlight/Spatio-Temporal-Prediction-Transformer-Review>.

CCS Concepts: • **General and reference** → **Surveys and overviews**; • **Computing methodologies** → **Artificial intelligence**;

Additional Key Words and Phrases: Spatio-temporal prediction, Transformer, foundation models, spatio-temporal correlations, AI, ML

ACM Reference Format:

Yingchi Mao, Hongliang Zhou, Ling Chen, Rongzhi Qi, Zhende Sun, Yi Rong, Xiaoming He, Mingkai Chen, Shahid Mumtaz, Valerio Frascolla, Mohsen Guizani, and Joel Rodrigues. 2025. A Survey on Spatio-Temporal Prediction: From Transformers to Foundation Models. *ACM Comput. Surv.* 58, 4, Article 85 (October 2025), 36 pages. <https://doi.org/10.1145/3766546>

1 Introduction

Spatio-temporal (ST) prediction is a significant field of application of **artificial intelligence (AI)**, focusing on modeling and predicting data with marked dynamic spatial and temporal characteristics. With an always growing number of deployed intelligent sensors and the pervasive uptake of the **Internet of Things (IoT)**, we are living in an era of big data with massive ST datasets[1]. Across a multitude of application domains, ranging from traffic flow [2] and weather forecast [3, 4], to human motion [5, 6], ST prediction plays a crucial role. Nevertheless, the intricacies inherent in ST data present significant challenges, as these datasets exhibit dynamic temporal sequences, spatial correlations, and complex nonlinear relationships. Traditional statistical and **machine learning (ML)** algorithms, while effective in certain contexts, often fall short in their ability to handle such nonlinear relationships. This shortcoming is primarily attributed to the underlying assumption of data independence inherent in these algorithms, which hampers their ability to effectively capture the spatial and temporal correlations amidst dynamic changes. In contrast, **deep learning (DL)** models, which offer greater expressive power and generalization capabilities, are better equipped to handle the complexity of ST prediction tasks. Currently, the DL methods that are widely used in the ST prediction field include **recurrent neural network (RNN)** [7, 8], **convolutional neural network (CNN)** [9, 10], **graph neural network (GNN)** [11, 12], and Transformers [13, 14]. These methods have been instrumental in advancing the field of ST prediction by providing more sophisticated and nuanced models, capable of capturing the complex relationships within ST data.

In recent years, with the advancement of DL techniques, Transformer-based methods have become a popular choice to handle the ST prediction tasks [15], as shown in Figure 1. The Transformer [16] architecture, initially conceived for sequence-to-sequence tasks, has been leveraging its innovative self-attention mechanism and prowess in managing long-range dependencies to demonstrate extraordinary potential in the realm of ST prediction. The self-attention mechanism facilitates the capture of both spatial and temporal dependencies, while learning the complex associations between these dimensions. This enhanced capability results in significant improvements in the accuracy of predictions. Furthermore, compared to RNN, the multi-head self-attention mechanism enables parallel processing of inputs, allowing Transformers to train and infer more efficiently on large-scale datasets. However, the complexity of the Transformer architecture, with a large number of parameters, necessitates big enough training data to avoid overfitting. The exponential growth in data volumes, the increased complexity of data, and more stringent prediction requirements have

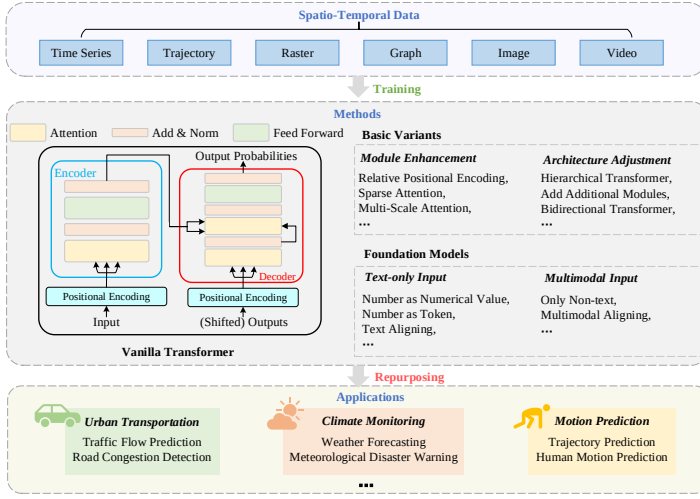


Fig. 1. ST data classification, prediction methods, and application domains.

further propelled the adoption of large Transformer models for ST prediction. Foundation models, characterized by their expansive parameters and sophisticated structures, have demonstrated an unparalleled ability to encapsulate the complex patterns embedded within ST data. Their robust adaptability and generalization capabilities are exemplified by their resilience in the face of shifting data distributions, as they continue to maintain superior prediction performance. This resilience positions them as ideal candidates for real-time or near-real-time prediction scenarios, where the ability to adapt to dynamic changes is of critical importance. In essence, Transformer-based methods, through their remarkable modeling capabilities and inherent flexibility, offer a superior framework for navigating the complex correlations encountered in ST prediction tasks.

In this article, we summarize Transformer-based prediction models for ST data from multiple perspectives. Especially, this survey focuses on studies that utilize Transformer-based models, with data that conforms to the definitions of ST data. We define Transformer models by their key feature, the attention mechanism; thus, any model that employs an attention mechanism is considered a derivative of the Transformer model. To the best of our knowledge, this is the first review to provide an in-depth introduction to ST prediction from various aspects, including architecture, applications, evaluation, and future directions. The main contributions of this survey are as following:

Harmonized and Structured Taxonomy. We introduce a uniformly structured taxonomy and classify ST prediction models based on the Transformer into three categories: module enhancement, architectural adjustment, and foundation models. Subsequently, we categorize and describe those three prediction models from the different application domains.

Abundant Resource Compilation. We curate a comprehensive array of public datasets across multiple application domains and evaluate various representative models, which are useful references for subsequent research in related fields.

Future Research Opportunities. Several promising research directions for future works are identified and detailed, encompassing a variety of aspects like low-quality ST prediction, model compression and real-time performance, model interpretability, model generalization capabilities, and external information fusion. This discussion offers readers a detailed understanding of the present advancements in ST prediction while highlighting forward-looking directions for future research.

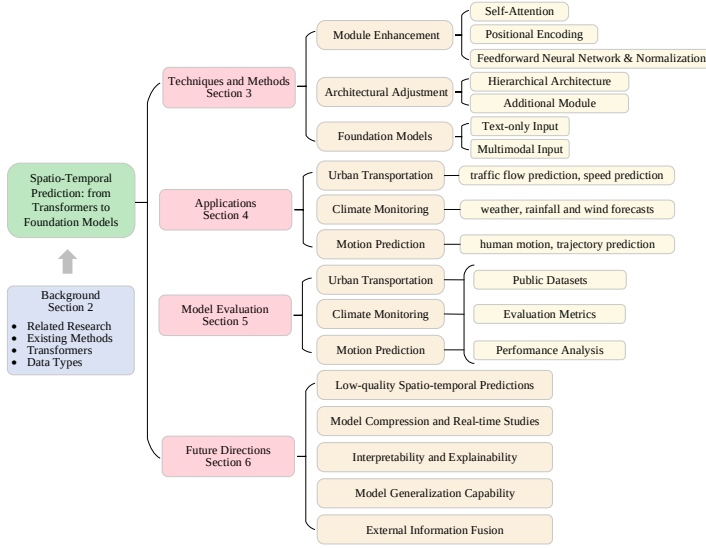


Fig. 2. A comprehensive taxonomy of Transformers for ST prediction in background, techniques and methods, application domains, model evaluation, and future directions.

The remainder of the article is structured as follows. Section 2 surveys the state-of-the-art and categorizes the existing methods, briefly introducing the model architecture and its Transformer variants. Section 3 delves into the main techniques and methods used by Transformer-based models. Section 4 demonstrates the application of Transformer models in ST prediction. Section 5 analyzes and evaluates typical models in different application domains. Section 6 identifies the limitations in ST prediction and outlines the future development directions of Transformer models. Conclusions are drawn in Section 7. Figure 2 illustrates a comprehensive taxonomy of Transformers for ST prediction in terms of background, techniques and methods, application domains, model evaluation, and future directions.

2 Background

2.1 State-of-the-Art Surveys

While several papers have explored the modeling of ST data through diverse methodologies, there has been a lack of focused attention on the application of Transformer-based methods for ST prediction. Atluri et al. [17] analyze prevalent challenges and techniques in ST data modeling, categorizing various forms of ST data to discern critical concerns within practical contexts. Wang et al. [9] provide an overview of recent progress in utilizing DL for ST data mining assignments and propose a comprehensive pipeline for DL models to address the intricacies of ST data modeling. Zhang et al. [18] conduct a survey on generative techniques for ST data and explore their application in ST data mining. To this end, they integrate ST generative methodologies and propose a standardized framework, which provides a systematic pipeline for selecting appropriate generative techniques within the realm of ST data mining. Wen et al. [19] provide a systematic review of time series analysis, focusing on the modeling of Transformers in tasks such as prediction, anomaly detection, and classification.

In the area of ST prediction, we can identify two main trends. The first one deals with methodological aspects, summarizing research findings under specific methods. For instance, Xie et al. [20] conduct a systematic examination of the urban traffic flow prediction challenge, grounded

in ST data, encapsulating a synopsis of prevailing ML-based methods, and succinctly address the obstacles and methodologies associated with urban traffic flow forecasting. Jin et al. [11] review the design methods of the **ST graph neural network (STGNN)** framework and its combination with other advanced technologies, summarize the limitations of current research and propose some prospective research directions. Jin et al. [21] offer a thorough review of large models for ST data, covering data formats, model classifications, model scopes, and application fields. Within the classification of **large models for ST data mining (LM4STD)**, they discuss **large language models (LLMs)** and pre-training foundation models for ST prediction.

The second trend focuses on practical application areas, summarizing prediction models across different fields. Ghoddusi et al. [22] provide a comprehensive survey concerning the utilization of ML methodologies within the domains of energy economics and finance. Their review encompasses a range of application contexts, such as the prognostication of electrical load, solar irradiance, wind energy generation, and financial market pricing. Their analysis reveals that **support vector machines (SVMs)**, **artificial neural networks (ANNs)**, and **genetic algorithms (GAs)** are the predominant algorithms employed in the energy economic sector. Liao et al. [23] briefly review attempts to use DL techniques in air quality prediction, introducing various deep network structures and highlighting their relevance in exploring nonlinear ST features of air pollution across multiple scales. Tedjopurnomo et al. [24] elaborate on the DNN architectures that are frequently utilized in research for traffic flow prediction. They systematically categorize these studies, highlight the shared features and distinctions among various methodologies, and elaborate on challenges and prospective research directions for future exploration in that domain. Badreddine et al. [25] conduct a review of scholarly works within the domain of crime prediction and organize these investigations into three distinct categories according to the employed methodologies: fundamental ML techniques, integrated ML approaches, and DL strategies. Their analysis indicates that **random forest (RF)** method is the prevailing choice in crime prediction, outperforming DL architectures. Zhang et al. [26] offer a comprehensive review of urban foundation models, which demonstrate a deep understanding of various urban data types and exhibit remarkable adaptability to a diverse range of urban tasks. These models represent a novel category of models that are pre-trained on extensive and varied urban data sources, and are adapted for a variety of downstream urban applications, including traffic flow management, energy management, environmental monitoring, and public safety and security. Zhang et al. [27] approach the topic of **point of interest (POI)** recommendation by utilizing significant amounts of ST data generated from smart devices and social networks to enhance POI recommendation systems. In addition, they also examine the growing concerns regarding information leakage and privacy preservation.

Although the aforementioned literature discusses the Transformer model, it does not fully demonstrate the potential of Transformer in ST prediction. We have created a comparison table to illustrate the unique contributions of this survey compared to existing research. As shown in Table 1, our article offers a broader coverage of the matter in focus, touching on all key aspects relevant in this field. The goal of this article is to provide a comprehensive and systematic review of Transformer-based methods in the domain of ST prediction. We intend to compare and analyze the relevant methods proposed in literature from the perspectives of model architecture, application areas, and experimental evaluations. The aim is to assist researchers in gaining a deeper understanding of the development and future directions of Transformer in ST prediction.

2.2 Classification of Existing Methods

2.2.1 Statistics and Machine Learning. In the early stages of ST prediction, researchers primarily relied on traditional time series forecasting and spatial interpolation methods. The most representative of these is the **autoregressive integrated moving average (ARIMA)** model [28]. Additionally,

Table 1. Comparison of Related Surveys

Survey	Year	Data		Models		Task
		Text	Multimodality	Transformers	Foundation Models	
[17]	2018	✓	✓	×	×	Multitask
[9]	2022	✓	✓	×	×	Multitask
[18]	2024	✓	✓	✓	✓	Generation
[19]	2023	✓	×	✓	×	Multitask
[20]	2020	✓	✓	×	×	Prediction
[11]	2024	✓	×	✓	×	Prediction
[21]	2023	✓	✓	×	✓	Multitask
[22]	2019	✓	×	×	×	Multitask
[23]	2020	✓	✓	×	×	Prediction
[24]	2022	✓	×	×	×	Prediction
[25]	2023	✓	✓	×	×	Prediction
[26]	2024	✓	✓	×	✓	Multitask
[27]	2025	✓	×	✓	✓	Recommendation
Ours	2025	✓	✓	✓	✓	Prediction

statistical methods such as **geographically weighted regression (GWR)** [29], **spatiotemporal kriging (ST-kriging)** [30], and **Bayesian maximum entropy (BME)** [31, 32] are also commonly used in the field of ST prediction. The ARIMA method emphasizes the analysis of historical trends and seasonality in time series data and is suitable for relatively simple linear ST relationships. Geostatistical methods like GWR and ST-kriging rely on spatial distance matrices and struggle to capture complex ST dependencies. For the complex ST dynamics in the real world, the effectiveness of these traditional statistical methods is somewhat limited. With the rise of ML techniques at the end of the 20th century, researchers began to apply these methods to ST prediction. ML algorithms such as RF [33], SVM [34], **linear regression (LR)** [35], **Bayesian networks (BN)** [36], and **multi-layer perceptrons (MLP)** [37] have attracted attention. However, one of the key challenges in this stage is that ML methods have relatively limited modeling capabilities for high-dimensional and dynamically changing ST data, especially when considering long-term dependencies and nonlinear relationships.

2.2.2 Deep Learning. Since the 2010s, with the rise of DL technology, ST prediction has undergone a revolutionary transformation. DL models, including CNNs and RNNs, have been introduced to augment the effectiveness of models in extracting intricate patterns within ST data. RNNs are suitable for processing temporal data, while CNNs can effectively extract spatial information. Research achievements in this phase have demonstrated remarkable performance in domains like traffic flow and meteorological changes. However, when tackling long-range dependencies, these models may face issues like vanishing or exploding gradients. In 2017, the emergence of the Transformer marked a new stage of development in ST prediction. The Transformer architecture has garnered considerable acclaim in language translation [38, 39], image processing [40, 41], and ST sequence prediction [42], effectively addressing the limitations of conventional DL approaches in capturing extensive-range ST dependencies. Particularly, it is noteworthy that GNNs have begun to emerge due to the availability of graph-structured ST data in some application domains, such as urban transportation networks [43] and social networks [44]. By modeling ST data as graphs, with nodes standing for locations or time points and edges representing ST relationships, GNNs can better capture the intricate associations between nodes, thus enhancing the precision of predictions [45]. For ST prediction without the graph structures, the Transformer is more suitable for capturing

global and long-distance dependencies, enabling the model to adapt to ST relationships across varying scales and temporal spans.

2.2.3 Integration Approach. In recent studies, researchers have attempted to synthesize fusion strategies of multiple methods to help overcome the limitations of individual approaches, thereby enhancing the predictive performance of models and making them more suitable for complex ST prediction tasks. Through the integration of diverse methodologies, a model can be enhanced in its ability to accommodate the variability and intricacy of the data, thereby augmenting the overall robustness. Some common DL fusion methods are: (1) CNN+RNN, where CNNs are responsible for extracting spatial features, while RNNs handle temporal information, jointly enhancing the modeling effect of ST data. Chen et al. [46] combine CNNs with **long short-term memory (LSTM)** networks [47] to propose a multi-factor ST wind speed prediction model. This model employs CNNs to discern spatial feature relationships among meteorological variables across various stations, and LSTMs to capture temporal feature relationships between consecutive historical time points. (2) GNN+RNN, where GNNs are designed to capture relationships between nodes, while RNNs process temporal information. Geng et al. [48] propose an ST correlation GNN for multi-node offshore wind speed prediction. This model uses a **graph convolutional network (GCN)** to replace traditional convolutions for Laplacian transformation, capturing potential spatial dependencies, and LSTMs to extract temporal correlations from high-dimensional spatial features. (3) GNN+Transformer, suitable for graph-structured ST data, where GNNs can capture relationships between nodes, while Transformers process global information.

2.2.4 Foundation Models. As mentioned above, researchers identified certain limitations of the Transformer model in handling multi-scale ST features and integrating external information. To address these issues, researchers began to explore more advanced models, i.e., foundation models. Foundation models, as general-purpose pre-trained models, acquire extensive prior knowledge by training on copious data, so to rapidly adapt to diverse downstream tasks. In 2018, the introduction of the **Bidirectional Encoder Representations from Transformers (BERT)** [49] model ushered in a new era of foundation model pre-training. BERT learns rich language representations by training on extensive text corpora and then fine-tuning various downstream tasks, resulting in significant performance improvements. Subsequently, a series of pre-training models based on the Transformer architecture emerged, like GPT [50], LLaMA [51], and GPT-4o [52], having the number of model parameters continuously increasing allows them to better capture intricate relationships and patterns in the data. As large-scale pre-trained models such as GPT gained popularity, researchers began to explore their application in ST prediction. These models, first pre-trained on large datasets, are subsequently fine-tuned for specific ST prediction tasks to enhance prediction accuracy and generalization. A new method was proposed by authors in [53] that employs a decoder architecture akin to that of LLMs and employs an autoregressive generation pre-training mechanism to address a spectrum of time series analysis objectives such as prediction, interpolation, and the detection of anomalies. Following pre-training, this method is further refined through fine-tuning to suit various specialized downstream applications.

2.3 Dominance of Transformers

Initially, the Transformer was proposed as a translation model and achieved state-of-the-art results at the time. As the Transformer architecture found wide application in various fields, researchers also began to explore its utility in ST prediction. A Vanilla Transformer first maps the input sequence into vector representations through an embedding layer and adds positional encoding to supply the model with temporal positional data. Subsequently, the obtained vector representations are input into the encoder-decoder architecture. The encoder is combined with several stacked

encoder blocks, each utilizing multi-head self-attention to seize dependencies within the sequence. Residual connections and normalization layers are added to accelerate training and produce context vectors. Meanwhile, the decoder structure echoes that of the encoder, with each decoder block incorporating a masked multi-head self-attention mechanism and residual connections accompanied by normalization layers. Each step of attention in the decoder block focuses on the encoder output and its previous prediction to generate the prediction sequence. Finally, the predicted results, after being processed through the decoder, are applied to the embedding layer to be transformed back into the original space.

The self-attention mechanism in Transformers models the relevance between elements within an input sequence, enabling simultaneous capture of spatial and temporal dependencies. Spatially, it can identify long-range relationships between different regions, overcoming the limitations of local receptive fields typical of CNNs. Temporally, it allows direct connections between features at arbitrary timesteps, facilitating the recognition of distant events or state changes. Compared to traditional CNNs and RNNs, Transformers excel in modeling long-range dependencies and offer superior parallel processing capabilities, especially when dealing with large-scale datasets. Since RNNs are hard to perform parallel computation on long sequences and CNNs require additional modules to capture long-range dependencies, the attention mechanism demonstrates superior computational efficiency. Consequently, the proven advantages of Transformers in NLP and computer vision, such as effective long-distance dependency modeling and efficient sparse attention computing, have been successfully extended to ST tasks, resulting in more comprehensive understanding and representation of complex space-time data. Consequently, Transformers have gradually become the mainstream approach in the field of ST prediction, leading to numerous variants.

As the available datasets have grown larger and GPU computing resources have become more abundant, OpenAI launched the renowned GPT-3.5 based on the Transformer architecture, achieving remarkable success in natural language communication. This has prompted increased attention toward foundation models, with efforts to enhance the predictive accuracy of ST predictions. Currently, foundation models have shown promise in this domain, particularly excelling in few-shot and zero-shot tasks.

2.4 Spatio-Temporal Data Types

ST data refer to datasets that integrate temporal (time-based) and spatial (location-based) information, capturing dynamic patterns and relationships over time and space. Specifically, temporal information refers to data that embodies temporal relationships, which can manifest as data with timestamps or a clear sequence, such as time series, trajectories, and sequential images and videos. In contrast, spatial information encompasses a wide array of spatial relationships, including location data, images, and various types of data that can explicitly or implicitly capture spatial relationships, such as time series, trajectories, ST grids, and ST graphs. The basic ST data structures discussed in this survey include time series, locations, trajectories, ST rasters, ST graphs, images, videos, and their definitions are described as follows.

Definition 1 (Time Series). A univariate time series $x = \{x_1, x_2, \dots, x_T\} \in \mathbb{R}^T$ is a sequence of T data points indexed in time order, where $x_t \in \mathbb{R}$ is the value of the time series at time t . A multivariate time series $X = \{X_1, X_2, \dots, X_T\} \in \mathbb{R}^{T \times D}$ is a sequence of T data points indexed in time order but with D dimensions, where $X_t \in \mathbb{R}^D (1 \leq t \leq T)$ denotes the values of the time series at time t along D channels.

Definition 2 (Location). A location refers to a fixed spatial point or object in a geographical space, represented by the geospatial coordinates $l \in \mathbb{R}^2$, i.e., latitude and longitude. It is often profiled by the corresponding satellite image, street-view image, and descriptions.

Definition 3 (Trajectory). A trajectory is a sequence of time-ordered locations that describe the movements of an object in the geographical space. It can be formulated as $\mathcal{T} = p_1 \rightarrow p_2 \rightarrow \dots \rightarrow p_T$, where $p_i = (l_i, t_i)$, and l_i denotes the location of an object at time t_i .

Definition 4 (ST Raster). An ST raster can be denoted as $\mathcal{X} = \langle X_1, X_2, \dots, X_T \rangle \in \mathbb{R}^{H \times W \times T \times D}$, where $X_T \in \mathbb{R}^{H \times W \times D}$ is the signals collected from $N = HW$ evenly distributed locations at time t , each characterized by D feature attributes.

Definition 5 (ST Graph). An ST graph extends the ST raster to be $\mathcal{X} = \langle X_1, X_2, \dots, X_T \rangle \in \mathbb{R}^{N \times T \times D}$ by explicitly incorporating spatial correlations with a graph $\mathcal{G}_t = (V, E_t, A_t)$ when N locations are not uniformly distributed, where V is the set of nodes, E_t is the set of edges, and $A_t \in \mathbb{R}^{N \times N}$ is the adjacency matrix at time t . The size of V is usually static. Furthermore, ST graphs can also be implicitly derived from time series data that exhibit spatial relationships.

Definition 6 (Image). Image F is a matrix of pixels, i.e., $F \in \mathbb{R}^{H \times W \times C}$, without considering other metadata, where H, W , and C are the height, width, and color channels of the image, respectively.

Definition 7 (Video). Let $\mathcal{V} = \{F_1, F_2, \dots, F_T\}$ be a video consisting of T frames indexed in time order, where F_t denotes the t -th frame. For simplicity, each frame F_t is a matrix of pixels, i.e., $F \in \mathbb{R}^{H \times W \times C}$, without considering other metadata, where H, W , and C are the height, width, and color channels of the frame, respectively.

3 Exploring Transformers: Techniques and Methods

Since its inception, the Transformer has achieved notable performance in various tasks, and improvements are continuously added to the original model, mainly with the goal to enhance the effectiveness of the model in a broader range of scenarios. These variations primarily address limitations of the Transformer in aspects such as positional encoding, expressiveness of the self-attention module, training efficiency, model structure, and scale.

In this section, we discuss the improved Transformer models, which can be mainly categorized into three broad types: module enhancement, architectural adjustment, and foundation models, as depicted in Figure 3.

3.1 Module Enhancement

3.1.1 Self-Attention. Self-attention, the core component of the Transformer, holds a significant role in understanding and processing complex dependencies in ST data. Self-attention enables the model to process all elements in a sequence in parallel, making it not limited by the length of the sequence and effectively capturing long-range dependencies. Additionally, the self-attention mechanism globally calculates attention weights, considering all elements in the sequence simultaneously, thus better modeling global dependencies. The adaptable attention mechanisms enable the model to discount irrelevant data and focus on salient ST features. The self-attention framework entails three core concepts: Query, Key, and Value, all derived from the representations of the input sequence. The Query serves to correlate with all Keys, assessing the relevance or influence of each element within the sequence on the current element. The Key is used to compare with Query to determine the importance of each element in the sequence to the present element. The Value contains information about each element in the sequence. Once the importance of each element is determined through Query and Key, the Value is weighted and summed to produce the final output representation.

However, the attention mechanism normally requires a significant computational effort and has a time complexity of $O(N^2d)$, where N is length of the input sequence and d the dimension of the hidden layer. Additionally, the sequence position information obtained by the attention mechanism,

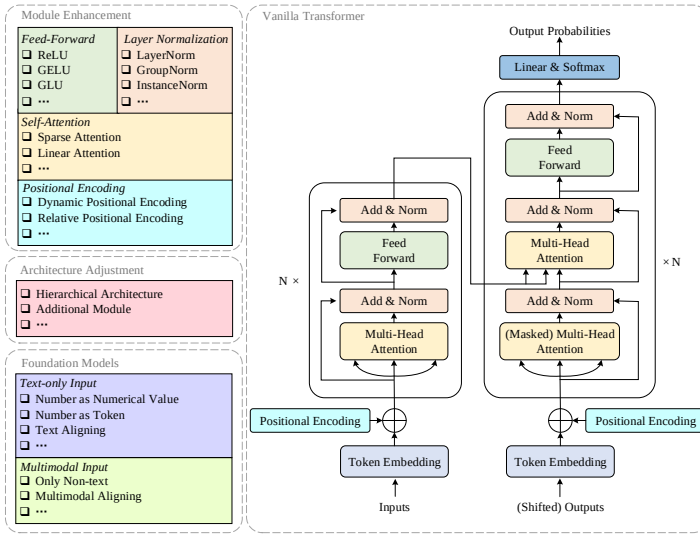


Fig. 3. The architecture of Vanilla Transformer and its main variants.

which is learned from training data, is prone to overfitting on small or medium-sized datasets [54]. In order to improve the predictive power and efficiency of the model, various variants of self-attention have been designed to advance the capacity of the model to understand complex ST data.

(1) Sparse Attention. In the standard self-attention mechanism, each output unit attends to the whole input units, which results in a significant increase in computational complexity for long sequences or high-dimensional data. Sparse attention addresses this issue by limiting the attention to specific parts of the sequence, thus reducing the number of attention pairs that need to be calculated. The implementation of sparse attention can be categorized into three main types.

- Fixed pattern sparse attention, which selects elements to attend to using predefined patterns. For example, the sliding window attention proposed by CE-GWT [55] primarily focuses on nearby elements for each element.
- Learned pattern sparse attention, which determines which elements should have attention weights through learning. For example, Synthesizer [56] employs a method called random attention, where some Query and Key pairs are randomly selected to calculate attention weights.
- Hierarchical attention, which segments the sequence into multiple blocks or segments, initially calculating attention within each block before computing attention between different blocks.

(2) Linear Attention. By introducing certain approximations or constraints, linear attention significantly reduces the quadratic computational complexity and memory requirements of traditional self-attention, achieving linear growth in both aspects. In ST prediction, a common implementation method is to learn a dynamic attention pattern, known as adaptive attention, to decrease the number of attention weights that need to be calculated. This method focuses only on the input elements that are most important for the current output, instead of considering all elements. Another type of linear attention achieves linearization by reordering the operations in the attention calculation.

This avoids the explicit computation and storage of the attention matrix, directly implementing the linearization of attention.

(3) Multi-Head Attention. The multi-head attention splits the input sequence into multiple heads, each equipped with its own Query, Key, and Value matrices. By doing so, each head learns a different attention pattern. These patterns are subsequently combined to generate the final output, enhancing the ability of the model to expressively capture different types of information simultaneously. In ST prediction, the multi-head attention aids the model in simultaneously capturing various scales and types of information. For example capturing global versus local information, or short-term trends and long-term correlations, or ST dependencies.

3.1.2 Positional Encoding. The original positional encoding in the Transformer supplements each element in the sequence with an extra vector, containing positional information, e.g., the index or the frequency components of the position. These vectors consist of the sequence embedding vectors to form the final input vectors. Since the positional information in the original positional encoding is fixed, the model cannot learn dynamic information related to position during training, which may lead to a lack of flexibility when handling ST data. Additionally, the design of the original positional encoding is primarily used for monomodal sequence data, and for other forms of data (e.g., images, videos) or multimodal data, the positional encoding needs to be able to handle different modalities, limiting the applicability of the original positional encoding. To address these shortcomings of the original positional encoding, various improvement methods have been proposed, aiming at enhancing the efficacy and generalization abilities of the model, allowing it to better handle various types of data and tasks.

- Dynamic Positional Encoding. It allows the positional encoding to adjust dynamically according to the characteristics of the data. For example, learned positional encoding learns the parameters of the positional encoding during the training process, allowing it to adapt to different types of data. The positional encoding implementations in architectures like Bert and GPT are learnable, treating the positional encoding as a trainable parameter that is updated along with the training process.
- Relative Positional Encoding. It generates positional encoding by considering the relative positions of elements within the input sequence, which helps the model better grasp long-range dependencies. A novel model proposed by Song et al. [57] makes use of relative positional encoding, which creates a positional encoding by taking into account the relative position of an element within the input sequence.
- Hierarchical Positional Encoding. It primarily takes into consideration the diversity of the levels of features and dependencies, thus proposing that each level may require a different positional encoding.
- Multimodal Positional Encoding. When processing multimodal data, different modalities may require different positional encodings. Multimodal positional encoding is designed to adapt to the features and dependencies of different modalities.
- Attention-based Positional Encoding. In the Transformer, the integration of positional encoding with the attention mechanism can result in a more effective capture of long-range dependencies. For instance, AirFormer [58] integrates positional encoding into the attention module to enhance the performance of the model.

3.1.3 Feedforward Neural Network and Layer Normalization. The FFN in the original Transformer, composed of two fully connected layers, transforms and extracts features from the input via linear transformations and activation functions. FFN can be represented as

$$\text{FFN}(H') = \text{ReLU}(H'W^1 + b^1)W^2 + b^2 \quad (1)$$

where H' denotes the output of the preceding layer, W^1 , W^2 , b^1 and b^2 are trainable parameters. The FFN performs a nonlinear transformation of the elements at each position, intending to enhance the nonlinear representation of the model. While the Vanilla Transformer utilizes a **Rectified Linear Unit (ReLU)** [59] activation to achieve a nonlinear relationship between the two FFN layers, alternative activation functions have been investigated, e.g., DPRELU [60] and GLU [61].

The Layer normalization block processes the outputs of each NN layer, while residual connections facilitate the transfer of information across layers. These two mechanisms mutually enhance each other, collectively improving the training quality and generalization capability of the Transformer, thereby enabling the design of deeper structures in deep networks. The computational formulas for layer normalization and residual connections are as follows:

$$H' = \text{LayerNorm}(\text{SelfAttn}(X) + X) \quad (2)$$

$$H = \text{LayerNorm}(\text{FFN}(H') + H') \quad (3)$$

where $\text{SelfAttn}(\cdot)$ denotes the self-attention module, X the input to the self-attention mechanism, $\text{SelfAttn}(X) + X$ the residual connection, and $\text{LayerNorm}(\cdot)$ the layer normalization operation. A common improvement strategy for the normalization layer is to change the LayerNorm [62] to GroupNorm [63] or $\text{Maximum Norm Minimization}$ [64]. Additionally, DeepMR [65] utilizes learnable residual connections to replace layer normalization in Transformers, constructing deeper neural networks.

3.2 Architecture Adjustment

ST data typically exhibit properties such as locality, periodicity, and long-term dependencies. To better adapt to ST data of different scales and characteristics, researchers have proposed various architectural adjustment methods. These architectural adjustment methods could be classified into two broad types: the introduction of hierarchical architecture and the incorporation of supplementary modules.

3.2.1 Hierarchical Architecture. One approach to adapting Transformers to ST data involves introducing a hierarchical architecture and decomposing the input vectors into finer-grained elements in a layered manner. In this methodology, primitive features are first introduced into the Transformer encoder to generate output representations. These representations are then aggregated to construct higher-order features, which are further processed by a superior-level Transformer. This hierarchical modeling strategy facilitates the ability of the model to handle extended inputs with constrained resources and may result in more nuanced representations advantageous for subsequent tasks. Informer [66] reduces the temporal dimensionality of features at a single level by inserting maximum pooling layers with varying spans between attention blocks, thereby enabling the model to process longer sequence inputs. Deepnet [67] develops intra-scale and inter-scale attention mechanisms to effectively grasp temporal dependencies across varying resolutions.

3.2.2 Additional Module. Another approach to adapting Transformers for ST data involves incorporating additional modules to enhance their ability to learn ST dependencies among different variables. Since the original Transformer architecture could not effectively handle spatial structure information, some works have added additional modules to learn ST dependencies among different variables. As a widely recognized network structure capable of handling spatial dependencies among different variables, GNN is integrated into Transformers. Depending on the depth of integration with Transformers, these approaches can be categorized into two main types: the first type involves using GNN as modules, directly incorporating them into the Transformer architecture; the second type further integrates attention mechanisms within GNN, resulting in a deeper fusion of GNN and Transformer.

- GNN as Modules. Given the limitations of Transformers in handling spatial dependencies, it is beneficial to assign the role of processing spatial dependencies to GNN while letting the Transformer handle temporal dependencies. GraphTrans [68] integrates a Transformer subnetwork with a standard GNN layer, the GNN layer specializes in learning local representations of an immediate neighborhood structure of a node, while the Transformer subnetwork computes all pairwise node interactions in a position-agnostic manner. A non-stationary GNNCrossformer [69] employs a non-stationary two-stage attention mechanism to address non-stationary cross-time dependencies and cross-dimensional dependencies. For on-stationary cross-time dependencies, it utilizes a graph CNN with Chebyshev interpolation to efficiently extract temporally conditioned topological information.
- GNN with Attention. When calculating the correlations between nodes, the attention mechanism is used to replace the traditional computation methods. For graph-structured data, the introduction of **graph Transformer networks (GTN)** [70] achieves the first application of Transformer in modeling graph-structured data. Building upon this concept, Graphformers [71] integrates GNN and Transformer by adding encodings that represent graph structural information in the process of calculating attention scores in the self-attention module, endowing the Transformer with the capability to encode graph structures. Furthermore, Crossformer [72] designs a two-stage attention mechanism, which includes both cross-temporal and cross-dimension attention mechanisms, used to capture temporal and inter-variable dependencies.

3.3 Foundation Models

This category refers to larger Transformer models that enhance the expressiveness of a model by increasing its width (more attention heads or hidden units) or depth (more layers). These foundation models are generally equipped to process extensive volumes of ST data and capture complex patterns and long-term dependencies within the data. Recent studies have found that pre-training Transformer models on large datasets allow them to develop general representations that prove advantageous for downstream tasks. These models undergo pre-training using a variety of self-supervised objectives. Following pre-training, simply fine-tuning these models on downstream datasets leads to improved experimental outcomes. For instance, BERT, a typical pre-trained model employing Only-Encoder architecture, is widely utilized for natural language understanding tasks. Foundation models, with their powerful computational capabilities and complex data processing abilities, have become a major trend in ST prediction. They could promote the efficiency and precision of data analysis and facilitate the discovery of valuable information across multiple domains, enhancing pattern recognition and reasoning capabilities across different fields.

Due to the fact that early foundation models only supported text as input, the main example being GPT-3.5, text-only foundation models have rapidly advanced in the field of ST forecasting. This progress has been driven by the large scale and easy accessibility of time series data, along with wide participation from researchers. Consequently, various input-based prompting strategies have emerged, which include treating time series as numerical data, converting time series into tokens, aligning time series with natural language, and so on. As foundation models evolve toward multimodality, new multimodal foundation models that utilize ST data from images and videos have emerged. These models can be categorized into two main types. One type consists of pure visual modality prediction models that use only images or videos as input. The other type includes multimodal fusion models, which enhance predictive performance through cross-modal alignment between text data and multimodal data.

3.3.1 Text-Only Input. Text-only foundation models are primarily employed in ST forecasting applications involving time series data. In consideration of the powerful capabilities for feature

extraction of foundation models, it becomes particularly important to find methods that enable these models to extract latent features more efficiently, which is primarily reflected in the construction of prompts. In traditional ML and DL models, time series data are typically processed as numerical data. However, given the impressive ability of foundation models to handle natural language, some researchers have sought to convert time series data into tokens as input. Furthermore, in some fields, time series and text can be aligned through loss functions to enhance predictive accuracy.

- Number as Numerical Value. Similar to traditional ML and DL models, time series data is often treated merely as numerical data. Garza et al. [73] introduce TimeGPT, the first foundation model specifically designed for time series data, which can generate accurate predictions across various domains without additional training. FPT [74] leverages a frozen pre-trained language model (GPT-2) to address all major time series task, achieving excellent results. Furthermore, TimesFM [75] explores a large pre-trained model trained on extensive time series data, demonstrating near state-of-the-art zero-shot accuracy when applied to a diverse array of previously unseen prediction datasets across different domains.
- Number as Token. One type of token converts numerical data into pre-defined text categories, which is primarily adopted in financial domain. For example, TDML [76] categorizes the weekly price fluctuations into 12 bins represented as D_i or U_i , where D indicates a decrease in price, U an increase, and $i = 1, 2, 3, 4, 5, 5+$ represents the level of price change. The other type of token is obtained through the embedding layer commonly used in DL and serves as input to the foundation model. Time-LLM [77] reprograms time series data into text prototypes as input to LLaMA-7B. This approach allows LLMs to be easily applied to general time series prediction tasks without any training. Due to the constraints of the No Free Lunch theorem, Time-LLM can only achieve an advantage in specific domains. However, it could provide natural language prompts such as domain expert knowledge and task instructions to augment input context.
- Text Aligning. Time series embeddings can be aligned with text embeddings through similarity matching, such as minimizing contrastive loss. For instance, based on contrastive learning, ETP [78] aligns **electrocardiography (ECG)** signals with textual reports. TEST [79] employs contrastive learning to produce time series embeddings that are aligned with text embeddings on an instance-wise, feature-wise, and text-prototype level. In addition to contrastive loss, other loss functions can also be used to achieve similarity matching. MTAM [80] utilizes various alignment techniques, such as Canonical Correlation Analysis and Wasserstein Distance, as loss functions to align **electroencephalography (EEG)** features with their corresponding language descriptions.

3.3.2 Multimodal Input. As foundational models continue to evolve toward multimodality, new multimodal foundation models have emerged. These models effectively leverage ST data from images and videos. In some fields, research often relies solely on images or videos as inputs, thereby minimizing interference from other modalities. At the same time, some researchers have attempted to convert time series data into images and use these temporal images as inputs, aiming to utilize the robust feature extraction capabilities of foundational models for predictive purposes. Additionally, some researchers explore the alignment of text with multimodal data, using textual information to guide the prediction of multimodal datasets.

- Only Non-text. In weather forecasting, atmospheric data can be transformed into images for research purposes. PanGu [81] injects deep networks with earth-specific prior knowledge, utilizing a 3D model to capture the relationships between atmospheric states at different pressure levels. FengWu [82] treats different atmospheric variables as different modalities and uses modality-specific encoders and a cross-modality Transformer to model the interactions

between these variables. Besides, Insight Miner [83] transforms time series into images through line plots and utilizes these images as input into a vision-language model to generate descriptions of time series trends.

- **Multimodal Aligning.** Aligning text prompts with data from other modalities may enhance prediction accuracy. ImageBind [84] leverages image-paired data to integrate six modalities—images, text, audio, depth, thermal, and **Inertial Measurement Unit (IMU)** time series—creating a joint embedding space that facilitates new emergent capabilities. Meanwhile, PandaGPT [85] further enhances the multimodal encoders from ImageBind by incorporating a foundation model, thereby enabling the system to follow visual and auditory instructions.

4 Application of Spatio-Temporal Transformer

In this section, we review Transformer models in the primary application domains of ST prediction and explicitly categorize the models in each domain according to their improvement modules. ST data, with their complex temporal and spatial dependencies, require special consideration in model design to effectively capture these relationships. Many models attempt to capture the complex dependencies in temporal data making use of the powerful representation capabilities of the Transformer. To better accommodate the unique characteristics of data in different domains, researchers have explored various Transformer variants and have sought to combine CNNs and RNNs with the expectation of optimizing predictive performance. In modeling spatial dependencies, to more accurately capture the spatial information within ST data, most models employ two main strategies: develop specific Transformer variants for capturing spatial dependencies, or use GNN techniques to learn feature representations of the nodes, capturing nonlinear relationships and patterns in spatial data. In addition to modeling temporal and spatial dependencies separately, some models design various ST fusion mechanisms to simultaneously capture both.

4.1 Urban Transportation

Cities are hubs of population concentration, with transportation serving as a crucial infrastructure that supports the efficient functioning of various urban activities. Carrying out the task of predicting ST data such as traffic flow, speed, and pedestrian volume can assist traffic management departments in anticipating potential road congestion hotspots and implementing timely dispatching controls.

Module Enhancement. Cai et al. [86] propose an end-to-end approach for coherent modeling of the ST dependencies in traffic data, introducing four new position encoding strategies: relative position encoding, global position encoding, periodic position encoding, and time series segments, for capturing the continuity and periodicity within the data. Yan et al. [87] enhance the encoding approach by using a global encoder in conjunction with a global-local decoder, enabling the extraction and subsequent fusion of both global and local spatial patterns. Lastjormer [88] proposes an ST joint attention mechanism within the Transformer architecture to effectively extract all dynamic correlations present in traffic data. This approach employs dot-product self-attention between sensors across multiple timesteps to seize dependencies while incorporating linear and convolutional self-attention mechanisms, to minimize computational requirements and simultaneously integrate local ST information. ASTTGN [89] designs an adaptive ST Transformer graph network that utilizes adaptive spatial and temporal Transformer modules to capture ST dependencies. It employs a gated mechanism aggregation module to fuse and selectively enhance ST features. Mu2ReST [42] designs a recursive ST Transformer architecture tailored for multi-resolution long-term ST prediction tasks. It does not consider any predefined graph structure in the spatial self-attention mechanism and does not use positional embedding. Instead, it employs lightweight ST attention to capture spatial and temporal dependencies. ASTTN [90] employs a graph-structured Laplacian encoding method and designs an adaptive graph ST Transformer architecture for traffic flow prediction. STCTN

Table 2. Summary of the Module Enhancement Methods for Urban Transportation Applications

Model Name	Model Feature	Advantages	Limitations
Traffic Transformer[86]	multiple positional encoding strategies	capturing global-local context temporal dependencies	no consideration of spatial correlation
Traffic Transformer[87]	a global-local decoder	integrating local and global information	lack of external information fusion
Lastjormer[88]	ST joint attention	linear attention	poor interpretability
ASTTGN[89]	adaptive ST Transformer	selective feature fusion and enhancement	poor generalizability
Mu2ReST[42]	no positional embedding used	model lightweight; structural flexibility	inadequate coverage of datasets
ASTTN[90]	Laplace encoding	efficient computational attention	poor interpretability
STCTN[91]	local-range convolutional attention	capturing global-local context temporal dependencies	poor generalizability
MDTNet[92]	area encoding, feature encoding	integration of multiscale information	poor generalizability
TFT[93]	ST cross attention	high computational efficiency	model needs to be lightweight

[91] proposes a local range convolutional attention mechanism that can simultaneously attend to both global and local contextual temporal dependencies at the sequence level. To better capture the ST dependencies of the data, some studies have achieved improved prediction performance by adding additional modules to the Transformer architecture. MDTNet [92] develops a multi-scale ST Transformer network, incorporating regional encoding (categorical ST expectations) and feature encoding (auxiliary tasks) into the positional encoding. TFT [93] integrates GNN and Transformer, proposing an ST cross-attention strategy for refining information. Table 2 summarizes the features, advantages, and limitations of the module enhancement methods.

Architecture Adjustment. ASTGCN [94] employs attention modules with identical structures to process short-term dependencies, daily periodic dependencies, and weekly periodic dependencies in parallel. TSTNet [95] uses a graph attention network to learn spatial feature aggregation for establishing spatial dependencies and applies a sequence-to-sequence based Transformer structure to establish temporal dependencies. MVSTT [96] designs a multi-view ST Transformer network that integrates ST representations and time series features from multiple perspectives through an ST Transformer. Bi-STAT [97] utilizes an encoder-decoder structure which incorporates an ST adaptive Transformer to represent spatial relationships and capture temporal patterns, and investigates ST correlations using an interleaving approach. The model also includes a **dynamic halting module (DHM)** to moderately allocate the computational burden for each task. HSTGCNT [98] designs a model that combines a **long time series Transformer (LTT)** network with a **hierarchical ST graph convolutional network (HSTGCN)**. The LTT, by encoding temporal position information, extracts long-term temporal relationships from raw traffic flow data, while the HSTGCN, through cascading one-dimensional and graph convolutions, captures short-term temporal and spatial relationships within the traffic data. TE-TCN [99] introduces an innovative augmented Transformer temporal convolutional network designed to encapsulate spatial, long-term, and short-term periodic dependencies, thereby enhancing the precision of traffic flow predictions. This approach addresses the inadequacies of current research in concurrently fulfilling the demands of both long-term and short-term prediction objectives. CorrSTN [100] proposes two well-designed ST representations: spatial correlation information (SCorr) and temporal correlation information (TCorr). Using SCorr, the authors introduce a dynamic GNN module to integrate relevant feature into the spatial dependencies, as well as a multi-head attention module to accurately model dynamic temporal dependencies. With TCorr, the model explores correlation patterns between different periodic data to identify the most relevant information. Table 3 summarizes the features, advantages, and limitations of the architecture adjustment methods.

Table 3. Summary of the Architecture Adjustment Methods for Urban Transportation Applications

Model Name	Model Feature	Advantages	Limitations
ASTGCN[94]	introducing layered attention	integration of short-term, daily and weekly cycle dependencies	lack of external information fusion
TSTNet[95]	introducing graph attention	aggregating spatial features using graph attention	poor interpretability
MVSTT[96]	multi-granularity ST Transformer	fusion of multi-granularity information	poor generalizability
Bi-STAT[97]	ST adaptive Transformer	high computational efficiency	poor generalizability
HSTGCNT[98]	introducing hierarchical ST map convolution	integrating local and global information	lack of model run efficiency evaluation
TE-TCN[99]	adding additional modules	meet long-term and short-term forecasts	lack of external information fusion
CorrSTN[100]	adding additional modules	considering variable correlation	complex model structure

Foundation Models. Some models aim to achieve stronger prediction performance by utilizing pre-training Transformer models, primarily adopting the Only-Encoder Transformer architecture. As an example, Duan et al. [101] put forth a **Pre-training Bidirectional Temporal Representation (PBTR)** model, founded upon the Transformer encoder, which can automatically model very long-term temporal dependencies. Notably, PBTR exhibits simplicity, scalability, and the versatility to be integrated with any other spatial component. Drawing inspiration from BERT, Bitimebert [102] utilizes key features of the architecture and employs a similar bidirectional Transformer structure to forecast overall traffic flow. Unlike traditional models, Bitimebert improves the generalization capabilities of the model by leveraging data from multiple roads during the pre-training phase. TFM [103] uses a graph structure and a dynamic graph generation algorithm to seize the behaviors of members in the traffic system, accurately predicting traffic outcomes in densely populated urban areas. PGT [104] designs a parameter-sharing scheme to overcome similar information across various ST features. Additionally, the design elaborates a lightweight prompt-tuning strategy to adapt to various features. Through pre-training and prompt tuning stages, PGT can adapt to specific target features by prompting the backbone model while maintaining the learned common sense to enhance the capture of specific ST features. STP-FPTN [105], based on the Transformer encoder architecture, utilizes unsupervised pre-training to learn complex ST pattern representations, splitting traffic flow data based on sensor dimensions instead of time dimensions, and uses the Transformer model to capture complex and long-distance ST correlations. Table 4 summarizes the features, advantages, and limitations of the foundation models.

4.2 Climate Monitoring

Meteorological forecasting is crucial for various industries and public health, with its accuracy being directly related to the interests and safety of a country or region. For instance, accurate precipitation forecasts can assist agricultural departments in planning and reducing losses, while wind speed predictions can issue advance warnings to ensure maritime transportation safety. Air quality forecasts can alert the public to take protective measures and reduce health risks. Macroscopically, climate change affects national economic operations and social development. Only by accurately understanding its patterns can we take responsive measures to achieve sustainable development of the economy and society. Existing climate monitoring models based on the Transformer are primarily categorized into the following types, as detailed in Table 5.

Module Enhancement. STGTN [106] designs an external attention mechanism to capture potential relationships between samples. AirFormer [58] designs a two-stage model that contains two novel self-attention mechanisms in the bottom-up deterministic stage. The model employs dartboard spatial self-attention (DS-MSA) to effectively capture spatial dependencies and causal temporal self-attention (CT-MSA) to capture temporal dependencies. In the top-down stochastic stage, it leverages latent variables to represent the inherent uncertainty present within air quality

Table 4. Summary of the Foundation Models for Urban Transportation Applications

Model Name	Model Feature	Advantages	Limitations
PBTR[101]	very long-term temporal dependence can be modeled	simple, scalable	inadequate coverage of datasets
Bitimebert[102]	introducing transfer learning	high generalization capacity	the amount of training data needs to be sufficiently adequate
TFM[103]	introducing motion graphics	high generalization capacity	the amount of training data needs to be sufficiently adequate
PGT[104]	model parameter sharing	model lightweight; high generalization capacity	the amount of training data needs to be sufficiently adequate
STP-FPTN[105]	unsupervised pre-trained augmented model	high computational efficiency; high generalization capacity	poor interpretability

Table 5. Summary of Models for Climate Monitoring Applications

Model Category	Model Name	Model Feature	Advantages	Limitations
Module Enhancement	STGTN[106]	designing external attention	ability to learn dynamic ST information	higher computational cost
	AirFormer[58]	introducing potential variables	prior knowledge of air pollution was considered	high computational complexity
	ST-Transformer[107]	introducing sparse attention	consideration of spatial, temporal and variable correlations	poor interpretability
	CSWinTT[108]	designing shifted window cross-attention	introducing cross-attention favors feature integration	poor interpretability
Architecture Adjustment	Earthformer[109]	introducing a layered encoder-decoder architecture	high generalization capacity	lack of external information fusion
	EEMD-Transformer[110]	introducing empirical modal decomposition	reducing data noise	poor interpretability
	Preformer[111]	designing the encoder-translator-decoder architecture	small number of participants	poor generalizability
	Mixformer[112]	applying seasonal trend decomposition	effectively capture complex cycles and trends	sensitive to data quality and integrity
Foundation Models	Storm Prediction[113]	introducing vision Transformer	model lightweight	reliance on large amounts of labeled data
	FML-ST[114]	proposing a federal learning approach	reduce communication and computation overhead	high modeling complexity
	ClimaX[115]	design coding and aggregation modules	versatility; scalability	lack of assessment of forecasts of extreme events
	FengWu[82]	multimodal, multitasking model	high computational efficiency	lack of quantification of model uncertainty

data. ST-Transformer [107] adopts a sparse attention mechanism that focuses on valuable and relevant context information across spatial, temporal, and variable dimensions, and improves the self-attention mechanism module in decoder blocks to a sparse encoding-decoding attention mechanism. CSWinTT [108] designs a shifted window cross-attention mechanism that simultaneously attends to features within local regions and global features that span window boundaries.

Architecture Adjustment. Earthformer [109] utilizes a hierarchical encoder-decoder architecture to encode input sequences into multiple levels of representation. Predictions are then generated through a process of refinement from coarse to fine granularity. EEMD-Transformer [110] is based on an encoder-decoder architecture. There are an encoder that generates representations of arbitrary-length historical wind speed sequences, and a decoder that generates arbitrary-length future wind speed sequences. To facilitate modeling, the homogeneous **ensemble empirical modal decomposition (EEMD)** converts one-dimensional raw wind speed sequences into 16-dimensional sequences and the Transformer is then able to directly model this multi-dimensional wind speed data. Preformer [111] implements an encoder-translator-decoder architecture, where the encoder is responsible for integrating spatial features from multiple elements, the translator models ST dynamics, and the decoder combines the ST information to predict future precipitation. Mixformer [112] designs a new hybrid Transformer, incorporating a hierarchical context, for the purpose of ST wind speed prediction. Mixformer first employs a seasonal trend decomposition, then predicts the trend component using MLP, and predicts the seasonal component using an attention model.

Additionally, Mixformer introduces a **spatio-temporal Gaussian mixture attention (ST-GMA)** layer that integrates periodic temporal and long-term spatial contexts.

Foundation Models. Lin et al. [113] develop a lightweight, portable model architecture based on a Transformer encoder. This model utilizes satellite images as inputs for storm prediction in real-world applications. The model architecture is comprised of two stages. In the first stage, referred to as the representation learning stage, multiple masked autoencoders with varying scalability are employed to extract image patterns at multiple scales. In the second stage, a Vision Transformer-based encoder [116] processes input sequences packaged by the input representation learning stage for storm prediction. FML-ST [114], to alleviate the issue of data exposure across regions, proposes a new federated learning approach to collaboratively learn a model based on a new ST Transformer encoder. The model uses participant data with heterogeneous meteorological data and adopts a novel prompt learning mechanism to meet communication and computational constraints of low-resource sensors. ClimaX [115] expands upon the Transformer architecture by incorporating new encoding and aggregation modules, allowing for the efficient use of available computations while ensuring general utility. FengWu [82] proposes a multimodal and multitasking model to improve medium-range weather forecasting. It suggests treating different atmospheric variables as different modalities and uses modality-specific encoders and a cross-modality Transformer to model the interactions between these variables.

4.3 Motion Prediction

The movement of individuals or objects follows certain regularities, and accurate predictions of their future positions can provide valuable support for decision-making. For example, trajectory prediction of objects is widely applied in the domains of robotics, autonomous transportation, and unmanned aerial vehicles. Human body behavioral prediction can ensure public health in medical, sports, and health monitoring domains. To achieve precise motion prediction, researchers have proposed numerous methods based on DL. Among them, the Transformer, because of its strengths in capturing sequential dependencies, has seen extensive use in a variety of motion prediction tasks.

Module Enhancement. In the area of trajectory prediction, TrajFormer [117] leverages location-based social networks and semantic location encoding of geographical information to train the Transformer precisely on latitude and longitude coordinates for predicting the next destination of a cab. PRetr [118] utilizes a decomposed attention module to extract dependencies from a multi-agent scene and decodes a set of learned object queries in parallel using an encoder-decoder Transformer network. By avoiding iterative conditioning, this non-autoregressive approach reduces the amount of time needed for training and testing computation. RNTrajRec [119] captures spatial features using CNNs and proposes positional encoders to encode spatial features and corresponding temporal features. With the help of an attention mechanism, graph attention network blocks, based on the encoded features, are then used to model the interactions between agents as well as agents and infrastructure. A Transformer network is ultimately implemented to simultaneously predict the trajectories of multiple agents. In the field of human motion prediction, Aksan et al. [120] design a novel ST Transformer architecture with two core mechanisms. One is a decoupled ST attention mechanism that considers the current and past motion states simultaneously in each joint embedding. The other is a fully autoregressive modeling approach that reduces the error accumulated over time. When predicting three-dimensional human motion, STCT [121] designs a cross-transformer method to coherently link the temporal and spatial features. It proposes a reconstruction loss function that introduces skeletal constraints of the human mesh into the predicted three-dimensional human joints to produce more realistic three-dimensional human motion. Table 6 summarizes the features, advantages, and limitations of the module enhancement methods.

Table 6. Summary of the Module Enhancement Methods for Motion Prediction Applications

Model Name	Model Feature	Advantages	Limitations
TrajFormer[117]	designing semantic positional encoding for geographic information	precise longitude and latitude coordinates	poor generalizability
PReTR[118]	using parallel decoding mechanisms	high computational efficiency	poor interpretability
RNTrajRec[119]	designing position encoders	using sparse graph attention	high training cost
ST-Transformer[120]	designing decoupled ST attention	applicable to long-term forecasting	poor interpretability
STCT[121]	designing cross-transformer	introducing a reconstruction loss function	high modeling complexity

Architecture Adjustment. In the area of trajectory prediction, Stndt [122] designs a Transformer architecture combining spatial and temporal dimensions, including two core components: a spatial self-attention sublayer to capture global spatial interactions among all traffic participants, and a temporal convolutional network to capture temporal relationships between consecutive frames. B-STAR [123] designs a DL framework for multi-drone trajectory prediction, using Bayesian methods in the decoder to improve model robustness and external graph memory to help the spatial Transformer adjust temporal embeddings. HINormer [124] models driving scenes as heterogeneous graph networks with nodes and edges. Nodes are connected according to different semantic relationships to realize the construction of heterogeneous graphs. The Transformer architecture in a hierarchical manner enables adaptation to the heteroid inputs. LSSTA [125] proposes a GNN-based spatial encoder to depict spatial interactions among pedestrians and uses the Transformer network to process long-term temporal dependencies. TUTR [126], an innovative approach that obviates the requirement for post-processing, integrates trajectory forecasting, social interaction modeling, and multimodal trajectory prediction within a unified Transformer encoder-decoder framework. In detail, TUTR delves into the interplay among diverse motion behaviors by employing explicit global prediction mechanisms and implicit pattern-level transformer encoders. It concentrates on capturing social dynamics with nearby agents via a dedicated social-level transformer decoder. Moreover, the dual prediction capability allows for the concurrent forecasting of multiple trajectory alternatives along with their respective probabilities, all within a framework that dispenses with the need for subsequent processing stages. HPNet [127] integrates map constraints into the network architecture by concatenating a map and reference extraction module, and introduces an innovative multi-task optimization strategy to facilitate the learning of topological and rule-based constraints. Furthermore, HPNet employs a bilateral query mechanism for context fusion, resulting in a more efficient and streamlined network design. For 2D vehicle trajectory prediction at urban intersections, STTN [128] employs a domain adaptation training strategy that encompasses two distinct modules. The initial module, trajectory manipulation characterization, is responsible for identifying potential driving manipulation features and categorizing trajectory data into various types according to their respective distributions. The subsequent module, trajectory distribution matching, utilizes a novel ST Transformer network equipped with a distribution matching loss function to facilitate the dynamic learning process and accomplish precise trajectory predictions. In the field of human action prediction, LPJP [129] employs the **discrete cosine transform (DCT)** to capture the temporal dependencies of individual joint as frequency coefficients. These DCT coefficients for the temporal nodes are then integrated into a Transformer-based framework. To capitalize on the structural connectivity within the human skeleton, the model adopts an incremental decoding strategy, which predicts future joint dynamics sequentially, starting from the central joints and progressing to the peripheral joints, in line with the kinematic chain in human anatomy. SCAFF [130] designs a new spatial-temporal network that combines Transformer and GCN technologies. It utilizes the temporal

Table 7. Summary of the Architecture Adjustment Methods for Motion Prediction Applications

Model Name	Model Feature	Advantages	Limitations
Stdnt[122]	introducing temporal convolution	high generalization capacity	high modeling complexity
B-STAR[123]	introduction of Bayesian methods	introducing external graph information	poor generalizability
HINormer[124]	hierarchical architecture Transformer	relative coordinates express node relations	poor interpretability
LSSTA[125]	combining scenario information	highly expandable	high computational complexity
TUTR[126]	multimodal model	providing probabilistic information	poor real-time performance
HPNet[127]	proposing a multitasking optimization strategy	model lightweight	poor generalizability
STTN[128]	introducing domain adaptive training	high generalization capacity	high modeling complexity
LPJP[129]	designing progressive decoding strategies	utilizing the structural connectivity of the human skeleton	poor generalizability
SCAFF[130]	introducing graph convolution	propose a forecast correction module	compressible network architecture

Transformer to grasp the global temporal relationships, and the spatial GCN component to identify the local spatial connections among human joints in every frame. To address the challenges of accumulating errors and discontinuities in predicting actions, it suggests a correction technique that merges the current predicted frame with the one before it. Table 7 summarizes the features, advantages, and limitations of the architecture adjustment methods.

Foundation Models. SGTN [131] designs a multimodal prediction Transformer architecture, which first converts pedestrians into pseudo-images using a social interaction ST graph, and then captures ST information through a GCN. Subsequently, the Transformer temporal extrapolator explores the patterns of pedestrian motion behavior and generates a probability distribution of future trajectories using a multimodal prediction module. STNet [132] designs a network architecture based on GNN and Transformer, using a conditional variational autoencoder as a preprocessor to predict pedestrians' destinations. The Social Transformer captures important social features affecting pedestrian motion using graph attention mechanisms and completes the prediction with past trajectories, predicted destinations, and current pedestrian position information. BEVGPT [133] designs a large generative pre-training model that combines forecasting driving scenes, making decisions, and planning movements. The model exclusively uses a **bird's-eye view (BEV)** as its input, which is fed into a generative pre-training Transformer to determine driving actions in response to the nearby traffic scenario. An optimization-driven method for motion planning is also crafted to guarantee that the driving path is both practical and smooth. Cross-modal TF [134] designs a DL model for improving pedestrian trajectory prediction performance, using a cross-modal feature fusion Transformer to seamlessly integrate multimodal features. The incorporation of a multi-head co-attention mechanism allows for the effective fusion of features, while the addition of an auxiliary self-supervised future prediction loss enhances the ability of the model to learn from both past and future multimodal features. Context-Aware [135] designs a context-aware multimodal Transformer architecture that fully extracts ST interaction features between pedestrian movement and the surrounding environment, utilizing a multimodal encoder. Afterward, a single-pass Transformer-based encoder-decoder structure generates the entire prediction result at once and realizes lightweight deployment on embedded edge devices. Table 8 summarizes the features, advantages, and limitations of the foundation models.

4.4 Others

In addition to the aforementioned domains, research on ST prediction models based on the Transformer can be also mentioned, among others, in energy, finance, and healthcare vertical markets. The Hybrid ST Transformer [136] designs a mixed CNN-Transformer model with three core components. The primary component is a CNN encoder that extracts high-level spatial features. The

Table 8. Summary of the Foundation Models for Motion Prediction Applications

Model Name	Model Feature	Advantages	Limitations
SGTN[131]	multimodal model	fusion of multimodal features	high modeling complexity
STNet[132]	designing a social feature extraction method	simple and efficient model structure	lack of external information fusion
BEVGPT[133]	input the bird's-eye view	inputs indicate simplicity	high computational complexity
Cross-modal TF[134]	multimodal model	fusion of multimodal features	poor real-time performance; high modeling complexity
Context-Aware[135]	multimodal model	fusion of multimodal features	lack of external information fusion

second component is a Transformer encoder that combines the spatial features extracted by the CNN with time series data. The third component is a CNN decoder that reconstructs the predictions through U-Net [137] style skip connections. Sun et al. [138] design an ST correlation Transformer architecture with two core mechanisms. The first mechanism assesses the quality of spatial information in wind speed measurements using the k-nearest neighbors and maximum information coefficient. The second mechanism integrates multi-dimensional information such as time, geographical coordinates, and wind turbine identity using feature vectors with positional embeddings to ensure that the model can capture the temporal evolution of wind conditions and geographical location characteristics. Chen et al. [139] present a novel framework that utilizes the graph structure inference capability of ChatGPT to enhance the prediction performance of GNNs. ChatGPT first extracts dynamic network architectures from news and incorporates the inferred network into GNNs to generate vector representations for downstream prediction tasks.

5 Model Evaluation

In this section, we offer a overview of the widely utilized public datasets relevant to urban transportation, climate monitoring, and motion prediction. Additionally, we have conducted a comparison of experimental outcomes from models built on the Transformer architecture within the realms of urban transportation and motion prediction, utilizing some of these public datasets. The evaluation of these models was carried out through an analysis of the experimental findings. Due to the specific differences in the application scenarios of models in the climate monitoring domain, we are unable to directly assess the performance of different models through experimental results. Consequently, we have conducted a detailed comparative analysis of the models from the perspectives of input features, output results, and innovations to evaluate their applicability and main advantages and disadvantages in different application scenarios.

5.1 Urban Transportation

5.1.1 Public Datasets. **Metro Traffic Los Angeles (METR-LA)** [140] is a common dataset used for traffic flow prediction, primarily used for research and evaluation of traffic prediction algorithm performance. The dataset contains traffic data from the Los Angeles area and can be utilized to predict traffic flow. The format of the METR-LA dataset is typically time series data, which includes traffic flow information from traffic sensors at different time points. Common data formats include timestamps, flow (indicating the traffic flow rate per unit time, usually represented as the count of vehicles detected by a sensor), and other features (including speed, density, etc.) to offer a detailed description of traffic conditions. The dataset contains data from 207 sensors from March to June 2012, with recordings made at 5-minute intervals, resulting in a total of 207 sensors and 34,272 time points.

Performance Measurement System (PeMS) [141] offers a consolidated database of traffic data gathered by the California Department of Transportation. It includes data from highways in California and datasets from other transportation companies and partner organizations in the state.

Table 9. Prediction Performance of Different Models on the METR-LA Dataset is Evaluated in the Metric MAE/RMSE/MAPE (%) (Optimal Performance is Indicated by Bolding, Suboptimal by Underlining)

Model	15 min	30 min	60 min
Traffic Transformer[87]	2.66/ <u>5.11</u> /6.75	3.00/6.06/ <u>8.00</u>	3.39/7.04/ <u>9.37</u>
Traffic Transformer[86]	2.43/4.73/6.57	2.79/ 5.61/7.45	<u>3.28</u> / 6.68 / 9.08
Lastjormer[88]	<u>2.64</u> / <u>5.11</u> / <u>6.74</u>	<u>2.99</u> / 6.01/8.13	3.36/7.03/9.67
ASTTGN[89]	2.67/5.19/6.83	3.03/6.10/8.34	3.49/7.25/9.79
ASTTN[90]	2.74/5.45/7.48	3.10/6.15/8.53	3.13 /7.10/9.84
Vanilla Transformer[16]	2.98/6.04/8.88	3.65/7.03/9.64	4.25/7.96/12.41

These datasets facilitate a unified and exhaustive evaluation of highway functionality, empowering users in making a well-informed operational choice by comprehending the current status of the highway system. They make it possible to identify possible solutions by analyzing traffic congestion points, which improves decision-making in general. The most widely used datasets are PEMS-BAY, PeMSD3, PeMSD4, PeMSD7, and PeMSD8. PEMS-BAY contains data from 325 detectors collected over a 6-month period (from 1 January 2017 to 31 May 2017), with a data interval of 5 minutes. PeMSD3 includes data collected every 5 minutes by 358 detectors from September to November 2018. PeMSD4 contains traffic flow data collected every 5 minutes by 307 detectors over a continuous 59-day period (from 1 January 2018 to 28 February 2018). PeMSD7 includes traffic data collected every 5 minutes by 883 detectors over a continuous 3-month period (from 1 May 2017 to 31 August 2017). PeMSD8 features data collected every 5 minutes by 170 detectors over a 62-day period (from 1 July 2016 to 31 August 2016).

TaxiNYC [142] offers driving records for taxis in New York City from 2011 to 2014, including pickup and drop-off times and locations. BikeNYC [143] comes from the New York City bike system covering the period from 1 April to 30 September 2014. It includes data on trip duration, numbers of the origin and destination stations, as well as the start and end times of each trip. TaxiSZ [144] provides GPS data for taxis collected on 156 streets in Shenzhen during the entire month of January 2015. Additionally, LargeST [145] is the first dataset applied to large-scale traffic prediction, containing 8,600 sensors and covering 5 years.

5.1.2 Evaluation Metrics. For assessing the predictive performance of different approaches in urban transportation forecasting, we employed three commonly accepted measures, i.e., **Mean Absolute Error (MAE)**, **Root Mean Squared Error (RMSE)**, and **Mean Absolute Percentage Error (MAPE)**.

5.1.3 Performance Analysis. We conduct prediction experiments on the METR-LA dataset for 15 minutes (3 steps), 30 minutes (6 steps), and 60 minutes (12 steps) for six models. Table 9 displays the comparative results of the predictions from different models. Overall, the Traffic Transformer [86] achieves the best prediction performance across multiple timesteps, followed by Lastjormer, which achieves suboptimal prediction performance within shorter timesteps. The high performance of the Traffic Transformer may be attributed to its design of multiple position encoding strategies, which can capture the local and global continuity and periodicity of time series. Lastjormer considers the correlation between different sensors and integrates it into the local ST information. Compared to other models, the design of these two models is more conducive to capturing dynamic dependencies in traffic data.

Additionally, we carry out experiments on the PeMS series dataset to predict traffic flow 60 minutes ahead using 10 distinct models. The results of these experiments are shown in Table 10.

Table 10. Prediction Performance of Different Models on the PeMS Dataset for 60 Minutes is Evaluated in the Metric MAE/RMSE/MAPE (%) (Optimal Performance is Indicated by Bolding, Suboptimal by Underlining)

Model	PeMSD3	PeMSD4	PeMSD7	PeMSD8
ASTGCN[94]	17.34/29.56/17.21	22.93/35.22/16.56	24.01/37.87/10.73	18.25/28.06/11.64
Traffic Transformer[86]	16.39/27.87/15.84	19.16/30.57/13.70	23.90/36.85/10.90	15.37/24.21/10.09
Bi-STAT[97]	15.29/27.54/15.19	18.74/ <u>30.31</u> /12.59	20.64/34.03/8.88	14.07/23.45/9.27
STP-FPTN[105]	<u>14.54</u> / 24.12 /14.43	18.37 / 30.25 /12.47	<u>19.56</u> / <u>32.33</u> /8.48	<u>13.78</u> / <u>23.24</u> /9.23
CorrSTN[100]	14.48 / <u>24.61</u> / <u>13.58</u>	<u>18.46</u> /30.91/ <u>12.21</u>	18.10 / 31.61 / 7.58	12.56 / 22.42 / <u>8.53</u>
STCTN[91]	16.89/28.02/15.75	22.53/35.09/15.21	24.24/37.52/10.12	17.15/26.63/10.93
TE-TCN[99]	15.50/24.96/ 11.96	19.03/31.01/ 10.60	21.30/34.86/ <u>8.37</u>	15.22/24.39/ 8.10
MVSTT[96]	15.36/26.40/15.70	19.54/31.22/12.96	21.09/34.38/9.17	15.92/25.24/10.42
Vanilla Transformer[16]	20.01/30.01/23.12	24.06/37.66/17.25	28.07/41.42/14.13	18.93/28.11/13.69

When examining the results, it is observed that CorrSTN achieves the best prediction results on almost all datasets. This can be attributed to its design, Scorr, which considers the correlation between different periodic data and accurately captures the interrelationships between the data, providing a reliable representation for assessing the correlation strength. STP-FPTN achieves suboptimal prediction performance on most metrics. STP-FPTN decomposes traffic flow data along the sensor dimension, enabling the model to quickly capture long-distance correlations in the data and uses unsupervised pre-training to enhance the extraction of ST features. Compared to other models, CorrSTN can effectively learn the mutual relationships between different sensors, while STP-FPTN enhances its ability to learn complex ST pattern representations through pre-training.

5.2 Climate Monitoring

5.2.1 Public Datasets. ERA5 [146] is the global atmospheric reanalysis dataset released by the **European Center for Medium-Range Weather Forecasts (ECMWF)**. This dataset provides high-resolution meteorological data and is frequently utilized in meteorology and climate research. ERA5 can provide data ranging from 1979 up to three months before the current time, including parameters such as temperature, precipitation, pressure, and wind speed.

The Shifts [147] dataset, procured from industrial and service sectors, encompasses three tasks, each associated with a unique data pattern. These data patterns include meteorological prediction, machine-based translation, and the anticipation of movement in autonomous driving. Real distributional changes impact all of these jobs and data patterns.

CPCB [148], which stands for Central Pollution Control Board, collects data from 703 air quality monitoring stations located in India, with data collection intervals of 15 minutes. CMPI6 [149] is a significant dataset created to accelerate global climate modeling and prediction research and to foster international collaboration in the study of global climate change. Developed and maintained by model developers and climate scientists worldwide, the CMIP6 dataset contains model output data from multiple climate models, used for analyzing global climate change and forecasting future climate trends.

5.2.2 Evaluation Metrics. In the domain of climate monitoring, the lack of uniformity in specific forecasting tasks makes it unfeasible to employ a universal set of evaluation metrics for experimental assessment.

5.2.3 Performance Analysis. Due to the different application scenarios and datasets of meteorological prediction models, directly comparing their prediction results becomes extremely challenging. To align the comparison as much as possible and highlight the similarities and dif-

Table 11. Model Structure, Innovations , and Parameters

Model	Model structure				Key innovations	Parameters
	End-to-end	Two-stage	Input formats	Output formats		
Earthformer[109]	✓		$N \times T \times H \times W$	$N \times \hat{T} \times H \times W$	Cuboid Attention	3.61M
AirFormer[58]		✓	$C \times T \times N$	$1 \times \hat{T} \times N$	Introducing Potential Random Variables	246K
ST-Transformer[107]	✓		$C \times T \times N$	$1 \times \hat{T} \times N$	Sparse Attention	742K
EEMD-Transformer[110]	✓		$1 \times T$	$1 \times \hat{T}$	Empirical modal decomposition	524K
Preformer[111]	✓		$B \times T \times C \times H \times W$ (B denotes batch size)	$B \times \hat{T} \times C \times H \times W$	Convolution and Transformer	5.82M
Storm Prediction[113]		✓	Image and event time description.	binary classification	Multi-scale encoders	11.3M
ClimaX[115]		✓	$N \times T \times H \times W$	$N \times \hat{T} \times H \times W$	Vision Transformer	7.76M
FengWu[82]	✓		$N \times T \times H \times W$	$N \times \hat{T} \times H \times W$	Multimodal encoders	4.53G

ferences among various models, we have summarized the overall structures and key innovations of typical models, presenting a macroscopic view of their differences and similarities in Table 11, where N expresses the total count of features, C the count of channels, T the historical timestep, $\hat{T}$ the predictive timestep, and $H \times W$ the spatial resolution. By categorizing the model structures and innovations, we can avoid unfair comparisons caused by differences in datasets and focus more on the capabilities of the models. In addition, we have also conducted a statistical analysis of the parameters for each model.

In weather forecasts, Earthformer designs end-to-end models with a unified input and output format, which is a matrix of spatial, temporal, and feature dimensions, while Earthformer introduces a new cuboid attention mechanism. Due to the complex cuboid attention mechanism and images as input, the overall model size reaches millions. In $PM_{2.5}$ concentration prediction, AirFormer and ST-Transformer have the same input and output format, and both have improved the attention mechanism to enhance prediction accuracy. However, ST-Transformer is an end-to-end model that explains the dependencies between space, time, and variables through the attention matrix, offering stronger interpretability. AirFormer, on the other hand, is a two-stage model that captures data uncertainty by introducing latent variables. In wind speed forecast, the EEMD-Transformer focuses on single-variable time series data and uses EEMD to decompose the original sequence data before capturing complex dependencies with the Transformer. In the context of time series data, both AirFormer and ST-Transformer improve the attention mechanism based on the Vanilla Transformer without significantly increasing the model scale. The EEMD-Transformer primarily utilizes EEMD to convert the original one-dimensional wind speed sequence into a 16-dimensional sequence, subsequently employing the Transformer to model this multidimensional wind speed data directly. As a result, the parameters of these three models remain relatively small. Preformer and Storm Prediction models are different from the other models in that they are only applicable to specific data inputs, for precipitation and storm forecasts, respectively. For complex meteorological prediction, ClimaX adopts a Vision Transformer structure and designs variable tokenization and variable aggregation modules to handle heterogeneous data, demonstrating the ability to generalize to unseen variables in downstream tasks. When dealing with image datasets, the model scale naturally increases, resulting in parameters exceeding 5M for all three models. FengWu adopts a multi-variable multitask learning approach, more accurately balancing the weights of different meteorological variables to provide more accurate forecasts. FengWu approaches global medium-range weather forecasting from a multi-modal and multi-task perspective, where the extensive training data and intricate model architecture lead to large-scale parameters, requiring up to 17

days of training on a cluster of 32 Nvidia A100 GPUs. Overall, these models have their advantages and shortcomings, and analyzing the similarities and differences in their structures and applicable scenarios helps to assess the applicability of the models in specific scenarios and provides a basis for model selection.

5.3 Motion Prediction

5.3.1 Public Datasets. In trajectory prediction, the ETH [150] dataset is mainly divided into two parts. One part is used for computer vision tasks with the dataset content consisting of pedestrian movement videos captured on a plane. The other part is mainly used for social behavior modeling, providing an overview perspective of group pedestrian movement trajectories. The UCY [151] dataset consists of real pedestrian trajectories captured from an aerial perspective and is composed of three sequences (Zara01, Zara02, and UCY). INTERACTION [152] describes the trajectories of vehicles and pedestrians at road intersections or junctions, containing interaction patterns between vehicles and pedestrians. The dataset includes four scenarios (roundabout, no traffic lights, traffic lights, and lane changing/merging) and data from 12 locations. HighD [153] is a large natural vehicle trajectory dataset from German highways, collected by drones at a total of six locations, including 11.5 hours of measurements and 110,000 vehicles, as well as 5,600 complete lane change records. TRAF [154] focuses on high-density traffic conditions, with the dataset consisting of 720p RGB videos, each frame containing approximately 13 motor vehicles, 5 pedestrians, and 2 bicycles. The Waymo [155] dataset encompasses around 12 million annotation boxes for LiDAR and roughly the same number for image annotations, producing close to 113,000 object trajectories from LiDAR and approximately 250,000 trajectories from images. This dataset is organized into 1,000 training sets and 150 testing sets.

The PoseTrack [156] dataset features an extensive collection of video sequences that depict a wide array of everyday activities, including walking, running, and talking, among others. It includes 1,356 video sequences, 46,000 annotated video frames, and 276,000 human pose annotations. The Penn Action [157] dataset contains approximately 2,000 video sequences, each containing an average of 15 image frames. The video sequences have different lengths and types of actions, offering a certain level of diversity.

5.3.2 Evaluation Metrics. To evaluate the prediction performance of human behavior estimation models and trajectory prediction models, we apply the evaluation metric **Mean Per Joint Position Error (MPJPE)** for the former, and **Average Displacement Error (ADE)**, and **Final Displacement Error (FDE)**, for the latter.

MPJPE aims to compute the mean L_2 distance between the predicted joint points and the corresponding true joint points. A smaller value indicates better results. Specifically, assuming that $m_{f,S}^{(f)}(i)$ is a function to calculate the coordinates of joint i of skeleton S at frame f , and $m_{gt,S}^{(f)}(i)$ is the true value of joint i , MPJPE is defined as

$$\text{MPJPE}(f, S) = \frac{1}{N_S} \sum_{i=1}^{N_S} \|m_{f,S}^{(f)}(i) - m_{gt,S}^{(f)}(i)\|_2 \quad (4)$$

where N_S is the number of joints in skeleton S .

ADE is employed to measure the mean Euclidean distance between the predicted value and the actual value across all timesteps, with lower values denoting better results.

FDE refers to the final Euclidean distance between the predicted trajectory and the actual trajectory at the last timestep, where lower values suggest improved performance.

Table 12. Prediction Performance Using MPJPE Metrics on the Human3.6M Dataset (Optimal Performance is Indicated by Bolding, Suboptimal by Underlining)

Model	Walking	Eating	Smoking	Discussion
	80/160/320/400	80/160/320/400	80/160/320/400	80/160/320/400
ST-Transformer[120]	8.9/15.5/32.1/38.5	9.4/21.1/ <u>36.4</u> / <u>42.3</u>	8.8/15.2/25.1/ 24.5	<u>7.9</u> /25.7/39.9/ <u>47.5</u>
STCT[121]	6.9 / 13.5 / 26.3 / 29.7	6.7 / 15.7 / 29.8 / 40.2	6.5 / 10.3 / 23.5 /27.7	7.5 / 20.3 / 36.7 / 43.7
LPJP[129]	<u>7.9</u> / <u>14.5</u> / <u>29.1</u> / <u>34.5</u>	<u>8.4</u> / <u>18.1</u> / <u>37.4</u> / <u>45.3</u>	<u>6.8</u> / <u>13.2</u> / <u>24.1</u> / <u>27.5</u>	8.3/ <u>21.7</u> /43.9/48.0
Vanilla Transformer[16]	11.7/19.7/38.4/47.5	13.2/25.0/43.7/50.5	10.4/18.2/27.1/26.9	12.9/32.5/46.8/53.1

Table 13. Pedestrian Trajectory Prediction Results (Optimal Performance is Indicated by Bolding, Suboptimal by Underlining)

Model	Evaluation metrics: ADE↓ / FDE↓ (in meters)					
	ETH	HOTEL	ZARA1	ZARA2	UNIV	AVERAGE
B-STAR[123]	0.44/0.72	0.24/0.43	0.26/0.54	0.25/0.41	0.36/0.68	0.31/0.56
LSSTA[125]	0.30/0.52	<u>0.12</u> /0.20	0.20/0.40	0.16/0.32	0.28/0.55	0.21/0.40
TUTR[126]	0.40/0.61	0.11 / <u>0.18</u>	0.18/0.34	<u>0.13</u> /0.25	0.23/0.42	0.21/ <u>0.36</u>
SGTN[131]	0.40/ <u>0.59</u>	0.14/ 0.16	<u>0.17</u> / <u>0.29</u>	0.14/ 0.17	<u>0.16</u> / 0.18	<u>0.20</u> / 0.28
STNet[132]	<u>0.33</u> /0.47	0.16/0.25	0.23/0.41	0.18/0.34	0.32/0.55	0.24/0.41
Cross-modal TF[134]	0.41/0.81	0.19/0.35	0.11 / 0.20	0.12 / <u>0.19</u>	0.14 / <u>0.22</u>	0.19 / <u>0.36</u>
PReTR[118]	0.82/1.55	0.30/0.56	0.42/0.89	0.35/0.73	0.62/1.23	0.50/0.99
Vanilla Transformer[16]	1.03/2.10	0.36/0.71	0.44/1.00	0.34/0.76	0.53/1.32	0.54/1.17

5.3.3 Performance Analysis. The predictive performance of ST-Transformer, STCT and LPJP are assessed on the Human3.6M dataset, employing the MPJPE metric for evaluation. The findings, presented in Table 12, reveal that STCT surpasses the performance of the other three methods. Upon examining the model structures, we find that STCT incorporates a cross-transformer module between spatial and temporal branches, enabling the model to learn the coherence between spatial and temporal features. In contrast, models like ST-Transformer primarily focus on learning spatial or temporal features without considering the interaction between these features. Additionally, the network structure of STCT is relatively simple, with a reconstruction loss function used to generate realistic human motion. However, other models, such as ST-Transformer, use human joint positions directly as the final representation, which results in deviations between the predicted outcomes and the actual human poses.

We evaluate trajectory prediction models using the commonly available public datasets ETH and UCY, which consist of pedestrian trajectory videos captured in real-world scenarios with high social interaction. These videos include five datasets (ETH, HOTEL, UNVI, ZARA1, and ZARA2) and four situations. ZARA1 and ZARA2 are gathered at various times from the same place. Every 0.4 seconds, a sample of the trajectory video data is taken to observe 8 frames (3.2 seconds) and forecast the next 12 frames (4.8 seconds). To evaluate the prediction performance of the model, we assess personnel trajectory prediction using ADE and FDE metrics. The results of comparing the model prediction performances are showed in Table 13.

By comparing the prediction accuracy of various models, it can be observed that Cross-modal TF and SGTN outperform other methods in the UNIV, ZARA1, and ZARA2 scenes. UNIV has the highest pedestrian density and the most complex scene among the five datasets, indicating that Cross-modal TF and SGTN can extract information in complex scenarios. In contrast, the HOTEL dataset primarily contains linear trajectory data with a relatively low pedestrian density compared to the other datasets. Using complex DL models like Cross-modal TF in such scenarios can lead to overfitting, leading to a decrease in prediction performance.

6 Future Directions

In this section, we explore the possible constraints of present studies and highlight five future research directions aimed at developing Transformer-based prediction models for ST data with stronger and more reliable prediction performance.

6.1 Low-Quality Spatio-Temporal Predictions

In practical applications, factors such as network failures and sensor device damage often result in stored data containing a large number of missing values or even storage errors, leading to low data quality. However, in literature the vast majority of experiments on ST prediction use already processed, high-quality public datasets, allowing the models to achieve better prediction results on these datasets. If low-quality data are used directly for prediction, the results of the model will be greatly compromised and may not meet the needs of actual production and daily life. Common data preprocessing methods (e.g., random interpolation, 3sigma principle) are ineffective in handling large-scale low-quality data. Some studies concentrate on low-quality data prediction. For example, Cui et al. [158] introduce a network architecture consisting of both bidirectional and unidirectional stacked LSTM, equipped with an input unit designed to identify missing values and assist traffic prediction. Zhong et al. [159] develop a heterogeneous ST prediction framework that employs graph structures for modeling the evolving dependencies in traffic flow and utilizes RNN to grasp the temporal correlations of individual road segments. During the looping process, missing values are calculated and embedded into the prediction framework, allowing for the use of incomplete historical data for traffic prediction. Although these works address the issue of low-quality data prediction, compared to the research on high-quality data, there is still considerable room for development in the area of low-quality ST prediction. Besides, one of the issues with low-quality data is that the presence of large amounts of missing, noisy, or erroneous data can distort the true data distribution. Therefore, research on **out-of-distribution (OOD)** scenarios may offer inspiration and insights for improving low-quality ST predictions.

6.2 Model Compression and Real-Time Studies

For certain ST sequence prediction problems (e.g., weather prediction in localized areas, wind speed), the real-time nature of the prediction is crucial. However, the vast amount of ST data can lead to larger model size, which directly impacts prediction efficiency and makes it difficult to meet real-time requirements. Moreover, deploying large Transformer models on edge and mobile devices, which have rather low memory and computational resources, is challenging. Therefore, we need to establish a lightweight model that can be updated in real-time, reducing model size through model compression techniques while maintaining prediction effectiveness. For example, the Pdformer [160] modifies the Transformer structure and proposes **long short-range attention (LSRA)** to replace the self-attention mechanism. LSRA splits the input into two segments: the first one employs the conventional self-attention mechanism for global information extraction, while the second one utilizes convolutional kernels for local information extraction. This approach enhances the computational efficiency of the model and yields superior outcome with reduced calculations, making it apter for scenarios with scarce resource availability. To alleviate the challenges of deploying Transformer models practically, low-rank approximation and structured pruning are typically used as mainstream methods. However, structured pruning often inadvertently removes expressive neurons at high sparsity rates, resulting in a significant decline in model performance. Low-rank approximation aims to compress expressive neurons, but since the weight matrices of Transformer models are often full-rank, this might destroy the diversity of neurons and impact the capacity of the model to express. To address the limitations of structured pruning and low-rank approximation, some methods [161] approximate the weight matrix by the sum of the low-rank and

sparse matrix, using low-rank approximation to prevent excessive pruning of expressive neurons and sparse approximation to enhance the diversity of the low-rank approximation. Furthermore, inspired by Deepseek, researchers may consider incorporating the **mixture of experts (MOE)** architecture. While this model has a large overall parameters, only a small subset of parameters is activated at any given time. This characteristic not only reduces the demand for computational resources but also enhances inference efficiency.

6.3 Interpretability and Explainability

Prediction models based on DL generally struggle with interpretability issues, and thus improving the interpretability of Transformer models in ST prediction holds considerable importance. The high model interpretability can enhance the prediction results credibility, helping users understand the basis on which the model learns, and facilitate the discovery and correction of potential model defects, identifying its strengths and weaknesses in different ST regions. Explanation results can analyze whether the ST dependency patterns learned by the model conform to common sense and track the response of the model to changes in data. To enhance the interpretability of Transformer models, the Relevance Rollout technique has been developed, which provides a new visualization method for Transformer models to reveal the decision-making process during task handling. Relevance Rollout can handle the correlations between different layers in Transformer models and highlight the differences between different categories, thus aiding users in better understanding the decision-making process of the model. Additionally, this approach can be integrated with current explainability methods (like Transformer Programs [162]) to further enhance the interpretability of the model. ICOT [163] establishes a mathematically interpretable deep network with a structure similar to that of a Transformer, generating token representations through iterative optimization to reduce the sparsity of the target. These token representations are then further used for downstream tasks. The authors propose a self-attention model, which uses statistical correlations between tokens to improve token representations. Moreover, a structure-seeking model is introduced to find low-dimensional structures within the input data, representing them as inputs to the network. These methods collectively form a more unified framework, achieving both mathematical interpretability and good practical performance. In subsequent research, it is possible to explore the integration of physical models with Transformer-based models, combining the advantages of data-driven and model-driven approaches to enhance the accuracy of physical process understanding and ST prediction.

6.4 Model Generalization Capability

Given the complexity and uncertainty of ST data, models are prone to overfitting on specific datasets and might find it challenging to adjust to new data and application scenarios. Overfitted models may experience a significant decline in predictive accuracy when handling data containing OOD instances. Only by enhancing their generalization capabilities can models perform stably and reliably in practical applications. Consequently, it is essential to investigate how well models generalize across various scenarios and domains and their applicability in novel contexts. To enhance the generalization ability of Transformers, large-scale pre-training techniques have been introduced to learn more rich and general ST representations. For example, NuwaDynamics [164] establishes a causal concept for ST prediction, intending to pinpoint causal areas within the data and grant the model capabilities for causal reasoning through a two-stage process. Specifically, upstream self-supervision is first utilized to identify causally important blocks, equip the model with generalization information, and meaningfully intervene on complementary trivial blocks to extrapolate possible test distributions. Afterward, the data are transferred to downstream tasks for targeted ST targeting, aiding the model in recognizing a wider array of potential distributions

and enhancing its causal perception capabilities. Data augmentation helps the model learn to ignore unimportant features and focus on more general, task-critical features, thereby improving its generalization capability. Additionally, multi-source and multi-task learning can promote the ability of the model to seize general features. By forcing the model to learn multiple related tasks simultaneously, it fuses data information from multiple angles and modalities, thus identifying broader common patterns and regularities.

6.5 External Information Fusion

ST data inherently contain incompleteness and noise, making it challenging for models to accurately capture the complex ST dependencies within ST sequences when learning solely from these data. However, we can utilize relevant external information to compensate for the shortcomings of the original data, which include, but are not limited to, physical rules, geographic location features, geomorphological data, climatic features, satellite remote sensing data. Physical rules can reflect the underlying physical mechanisms of ST data, geographical information can provide regional spatial data, topographic data can offer insights into terrain influence, climatic characteristics can inform about climatic environmental impacts, and satellite remote sensing data can facilitate regional change monitoring. These external information sources reflect the multifaceted factors contributing to the formation of ST data. Integrating these external information sources can assist the Transformer in better learning ST dependencies from multiple perspectives: firstly, by supplementing and correcting the shortcomings of the original data; secondly, by providing additional contextual information; and thirdly, by guiding the model to learn knowledge that conforms to physical laws. Future research should focus on how to effectively integrate such structured and unstructured external information into the Transformer framework. Potential methods include multimodal fusion, structured attention mechanisms, or customized neural structures. Additionally, it is necessary to explore how to utilize this prior knowledge for self-explanation to enhance the interpretability and credibility of the learning process and results of the model. Only by digging deeper into the correlation between data from different sources can the Transformer provide more stable and precise ST predictions in actual complex environments, thereby meeting the needs of intelligent decision-making.

7 Conclusion

This article presents an in-depth investigation and analysis of Transformer models in ST prediction, focusing on three main domains, i.e., traffic flow, weather forecast, and human motion. This study aims to provide a new research perspective for ST prediction by introducing an innovative classification method to systematically categorize existing models. We first review relevant surveys on ST prediction and categorize existing methods. Subsequently, we provide a brief introduction to the original Transformer and its variants and categorize existing Transformer models into three categories: module enhancement, architectural adjustment, and foundation models. Based on this classification, we introduce common technical approaches and methods used by various Transformer variants and summarize the representative methods in different categories applied in selected domains, comparing the model performance through experimental evaluation or architectural analysis. Finally we point out some future research directions.

References

- [1] Valerio Frasca et al. 2022. Lesson learnt and future of AI applied to manufacturing. In *Intelligent Edge-Embedded Technologies for Digitising Industry*. 207–240.
- [2] Haitao Yuan and Guoliang Li. 2021. A survey of traffic prediction: From spatio-temporal data to intelligent transportation. *Data Science and Engineering* 6, 1 (2021), 63–85.

- [3] Junzhong Ji, Jing He, et al. 2024. Spatio-temporal transformer network for weather forecasting. *IEEE Transactions on Big Data* (2024), 1–16.
- [4] Obinna Ogbonna Otuu. 2024. Investigating the dependability of weather forecast application: A netnographic study. In *Proceedings of the 2024 Australian Computer-Human Interaction Conference (OzCHI)*. ACM, 462–471.
- [5] Mohammad Samin Yasar et al. 2024. PoseTron: Enabling close-proximity human-robot collaboration through multi-human motion prediction. In *Proceedings of the 2024 ACM/IEEE International Conference on HRI*. ACM, 830–839.
- [6] Jiexin Wang, Yujie Zhou, et al. 2023. Spatio-temporal branching for motion prediction using motion increments. In *Proceedings of the 2023 ACM International Conference on Multimedia (MM)*. ACM, New York, NY, USA, 4290–4299.
- [7] Chen Qiu, Yanyan Zhang, et al. 2018. Spatio-temporal wireless traffic prediction with recurrent neural network. *IEEE Wireless Communications Letters* 7, 4 (2018), 554–557.
- [8] Tania Panayiotou, Maria Michalopoulou, and Georgios Ellinas. 2023. Survey on machine learning for traffic-driven service provisioning in optical networks. *IEEE Communications Surveys & Tutorials* 25, 2 (2023), 1412–1443.
- [9] Senzhang Wang, Jiannong Cao, and Philip S. Yu. 2022. Deep learning for spatio-temporal data mining: A survey. *IEEE Transactions on Knowledge and Data Engineering* 34, 8 (2022), 3681–3700.
- [10] Dasa Hemanth Kumar and Samundiswary Punniakodi. 2023. A comparative survey on paddy leaf diseases detection using convolutional neural network algorithms. In *Proceedings of the 2023 ICACIC*. IEEE, 1–5.
- [11] Guangyin Jin, Yuxuan Liang, et al. 2023. Spatio-temporal graph neural networks for predictive learning in urban computing: A survey. *IEEE Transactions on Knowledge and Data Engineering (TKDE)* (2023), 1–20.
- [12] Hongjie Chen and Hoda Eldardiry. 2024. Graph time-series modeling in deep learning: A survey. *ACM Transactions on Knowledge Discovery from Data (TKDD)* 18, 5 (2024), 1–35.
- [13] Yao Liang, Lang Qin, et al. 2024. Traffic state prediction based on spatio-temporal graph transformer network. In *Proceedings of the 2023 International Conference on ACAL*. ACM, 222–227.
- [14] Hangchen Liu, Zheng Dong, et al. 2023. Spatio-temporal adaptive embedding makes vanilla transformer SOTA for traffic forecasting. In *Proceedings of the 2023 ACM CIKM*. ACM, 4125–4129.
- [15] Salman Khan et al. 2022. Transformers in vision: A survey. *ACM Computing Surveys* 54, 10 (2022), 1–41.
- [16] Ashish Vaswani, Noam Shazeer, et al. 2017. Attention is all you need. In *Proceedings of the 2017 Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 30. Curran Associates, Inc.
- [17] Gowtham Atluri, Anuj Karpatne, and Vipin Kumar. 2018. Spatio-temporal data mining: A survey of problems and methods. *ACM Computing Surveys* 51, 4 (2018), 83:1–83:41.
- [18] Qianru Zhang, Haixin Wang, et al. 2024. A survey of generative techniques for spatial-temporal data mining. arXiv:2405.09592. Retrieved from <https://arxiv.org/abs/2405.09592>
- [19] Qingsong Wen, Tian Zhou, et al. 2023. Transformers in time series: A survey. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*. IJCAI, 6778–6786.
- [20] Peng Xie, Tianrui Li, et al. 2020. Urban flow prediction from spatiotemporal data using machine learning: A survey. *Information Fusion* 59 (2020), 1–12.
- [21] Ming Jin, Qingsong Wen, et al. 2023. Large models for time series and spatio-temporal data: A survey and outlook. arXiv:2310.10196. Retrieved from <https://arxiv.org/abs/2310.10196>
- [22] Hamed Ghodusi, Germán G. Creamer, and Nima Rafizadeh. 2019. Machine learning in energy economics and finance: A review. *Energy Economics* 81 (2019), 709–727.
- [23] Qi Liao, Mingming Zhu, et al. 2020. Deep learning for air quality forecasts: A review. *Current Pollution Reports* 6 (2020), 399–409.
- [24] David Alexander Tedjopurnomo, Zhifeng Bao, et al. 2023. A survey on modern deep neural network for traffic prediction: Trends, methods and challenges. In *Proceedings of the 2023 IEEE ICDE*. IEEE, 3795–3796.
- [25] EL Gougi Badreddine, Hassouni Larbi, et al. 2023. Spatio-temporal crime forecasting: Approaches, datasets, and comparative study. In *Proceedings of the 2023 International Conference on AI2SD*. Springer, 231–251.
- [26] Weijia Zhang, Jindong Han, et al. 2024. Urban foundation models: A survey. In *Proceedings of the 2024 ACM SIGKDD Conference on Knowledge Discovery and Data Mining*. ACM, 6633–6643.
- [27] Qianru Zhang, Peng Yang, et al. 2025. A survey on point-of-interest recommendation: Models, architectures, and security. arXiv:2410.02191. Retrieved from <https://arxiv.org/abs/2410.02191>
- [28] Yanfang Pan, Bote Liu, et al. 2024. Forecasting analysis of logistics freight volume based on ARIMA model. In *Proceedings of the 2023 International Conference on BDEIM*. ACM, 882–887.
- [29] Mohammad Reza Shahneh, Samet Oymak, and Amr Magdy. 2021. A-GWR: Fast and accurate geospatial inference via augmented geographically weighted regression. In *Proceedings of the 2021 ACM SIGSPATIAL*. ACM, 564–575.
- [30] Chuanpan Zheng, Xiaoliang Fan, et al. 2023. INCREASE: Inductive graph representation learning for spatio-temporal kriging. In *Proceedings of the 2023 ACM Web Conference (WWW)*. ACM, 673–683.
- [31] Mei Shan, Shuang Liang, et al. 2021. Spatial prediction of soil calcium carbonate content based on Bayesian maximum entropy using environmental variables. *Nutrient Cycling in Agroecosystems* 120 (2021), 17–30.

- [32] Dongxia Wu, Ruijia Niu, et al. 2023. Deep Bayesian active learning for accelerating stochastic simulation. In *Proceedings of the 2023 ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*. ACM, 2559–2569.
- [33] Bashir Alam, Azmat Hussain, and Muhammad Fayaz. 2024. An effective approach for air quality prediction in Bishkek based on machine learning techniques. In *Proceedings of the 2023 ICAAI*. ACM, 42–47.
- [34] Qi Yang, Xin Han, et al. 2022. Combined load prediction model of SVM-LSTM based on Markov chain error correction. In *Proceedings of the 2022 ICIEI*. ACM, 143–148.
- [35] Sai Li, T. Tony Cai, and Hongzhe Li. 2022. Transfer learning for high-dimensional linear regression: Prediction, estimation and minimax optimality. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 84, 1 (2022), 149–173.
- [36] Muzhe Guo, Muhao Guo, et al. 2024. Bayesian iterative prediction and lexical-based interpretation for disturbed Chinese sentence pair matching. In *Proceedings of the 2024 ACM on Web Conference (WWW)*. ACM, 4618–4629.
- [37] Vijay Ekambaram, Arindam Jati, et al. 2023. TSMixer: Lightweight MLP-mixer model for multivariate time series forecasting. In *Proceedings of the 2023 ACM SIGKDD*. 459–469.
- [38] Yachao Li, Junhui Li, et al. 2023. P-Transformer: Towards better document-to-document neural machine translation. *IEEE/ACM Transactions on Audio, Speech and Language Processing* 31 (2023), 3859–3870.
- [39] Yiqing Lin. 2024. Transformer-based a automatic scoring model for translation jobs. In *Proceedings of the 2023 International Conference on Machine Learning and Computer Application (ICMLCA)*. ACM, 413–417.
- [40] Zhijian Xu, Xingyue Guo, and Juan Wang. 2023. Lightweight multi-scale transformer for automatic skin lesions segmentation. In *Proceedings of the 2023 ICDIP*. ACM, Article 22, 8 pages.
- [41] Zhipeng Zhao, Jianjun Zhang, et al. 2023. Semi-supervised 3D medical image segmentation using transformer and CNN. In *Proceedings of the 2023 International Conference on Digital Image Processing (ICDIP)*. ACM, Article 28, 9 pages.
- [42] Hao Niu, Chuizheng Meng, et al. 2022. Mu2rest: Multi-resolution recursive spatio-temporal transformer for long-term prediction. In *Proceedings of the 2022 PAKDD*. Springer, 68–80.
- [43] Weiwei Jiang and Jiayun Luo. 2022. Graph neural network for traffic forecasting: A survey. *Expert Systems with Applications* 207 (2022), 117921.
- [44] Dongsheng Luo, Wei Cheng, et al. 2020. Parameterized explainer for graph neural network. In *Proceedings of the 2020 Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 33. Curran Associates, Inc., 19620–19631.
- [45] Feiyan Sun, Wenning Hao, et al. 2024. A survey on spatio-temporal series prediction with deep learning: Taxonomy, applications, and future directions. *Neural Computing and Applications* (2024), 1–25.
- [46] Yong Chen, Shuai Zhang, et al. 2019. Multifactor spatio-temporal correlation model based on a combination of convolutional neural network and long short-term memory neural network for wind speed forecasting. *Energy Conversion and Management* 185 (2019), 783–799.
- [47] Greg Van Houdt, Carlos Mosquera, and Gonzalo Nápoles. 2020. A review on the long short-term memory model. *Artificial Intelligence Review* 53, 8 (2020), 5929–5955.
- [48] Xiulin Geng, Lingyu Xu, et al. 2021. Graph optimization neural network with spatio-temporal correlation learning for multi-node offshore wind speed forecasting. *Renewable Energy* 180 (2021), 1014–1025.
- [49] Jacob Devlin, Ming-Wei Chang, Kenton et al. 2019. BERT: Pre-training of deep bidirectional transformers for language understanding. In *Proceedings of the 2019 Conference of the NAACL-HLT*, Vol. 1. ACL, 4171–4186.
- [50] Zhe Li, Zhangyang Gao, et al. 2024. General point model pretraining with autoencoding and autoregressive. In *Proceedings of the 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 20954–20964.
- [51] Hugo Touvron, Thibaut Lavril, et al. 2023. Llama: Open and efficient foundation language models. arXiv:2302.13971. Retrieved from <https://arxiv.org/abs/2302.13971>
- [52] Ning Zhu, Nan Zhang, et al. 2024. OpenAI's GPT-4o in surgical oncology: Revolutionary advances in generative artificial intelligence. *European Journal of Cancer* 206 (2024).
- [53] Yong Liu, Haoran Zhang, et al. 2024. Timer: Transformers for time series analysis at scale. arXiv:2402.02368. Retrieved from <https://arxiv.org/abs/2402.02368>
- [54] Yi Tay, Mostafa Dehghani, et al. 2022. Efficient transformers: A survey. *ACM Computing Surveys* 55, 6 (2022), 1–28.
- [55] Haoyu Tian and Xin Ma. 2023. Skeleton-based action recognition with channel enhanced local graph window transformer. In *Proceedings of the 2023 ICACIC*. ACM, 195–202.
- [56] Yi Tay, Dara Bahri, et al. 2021. Synthesizer: Rethinking self-attention for transformer models. In *Proceedings of the 2021 International Conference on Machine Learning (ICML)*. ACM, 10183–10192.
- [57] Zhiyi Song, Dipankar Chaki, et al. 2023. Positional encoding-based resident identification in multi-resident smart homes. *ACM Transactions on Internet Technology* 24, 1, Article 1 (2023), 27 pages.
- [58] Yuxuan Liang, Yutong Xia, et al. 2023. Airformer: Predicting nationwide air quality in China with transformers. In *Proceedings of the 2023 AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 37. AAAI, 14329–14337.
- [59] Xavier Glorot, Antoine Bordes, and Yoshua Bengio. 2011. Deep sparse rectifier neural networks. In *Proceedings of the 2011 International Conference on Artificial Intelligence and Statistics (AISTATS)*. JMLR, 315–323.

- [60] Kien Mai Ngoc, Donghun Yang, et al. 2021. DPRELU: Dynamic parametric rectified linear unit. In *Proceedings of the 2021 International Conference on Smart Media and Applications (SMA)*. ACM, 121–125.
- [61] Shaohua Liu, Yisu Wang, et al. 2022. An efficient spatial–temporal model based on gated linear units for trajectory prediction. *Neurocomputing* 492 (2022), 593–600.
- [62] Ruibin Xiong, Yunchang Yang, et al. 2020. On layer normalization in the transformer architecture. In *Proceedings of the 2020 International Conference on Machine Learning (ICML)*. PMLR, 10524–10533.
- [63] Yuxin Wu and Kaiming He. 2018. Group normalization. In *Proceedings of the 2018 ECCV*. Springer, 3–19.
- [64] Seonjae Lee, Myoung Hoon Lee, and Jun Moon. 2022. Maximum norm minimization: A single-policy multi-objective reinforcement learning to expansion of the pareto front. In *Proceedings of the 2022 ACM CIKM*. ACM, 1064–1073.
- [65] Shereen Elsayed and Lars Schmidt-Thieme. 2023. Deep multi-representation model for click-through rate prediction. In *Proceedings of the 2023 International Joint Conference on Neural Networks (IJCNN)*. IEEE, 1–9.
- [66] Haoyi Zhou, Shanghang Zhang, et al. 2021. Informer: Beyond efficient transformer for long sequence time-series forecasting. In *Proceedings of the 2021 AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 35. AAAI, 11106–11115.
- [67] Hongyu Wang, Shuming Ma, et al. 2024. DeepNet: Scaling transformers to 1,000 layers. *IEEE Transactions on Pattern Analysis and Machine Intelligence* (2024), 1–14.
- [68] Zhanghao Wu, Paras Jain, et al. 2021. Representing long-range context for graph neural networks with global attention. In *Proceedings of the 2021 NeurIPS*, Vol. 34. Curran Associates, Inc., 13266–13279.
- [69] Xinning Wu, Haolin Zhan, et al. 2025. Non-stationary GNNCrossformer: Transformer with graph information for non-stationary multivariate spatio-temporal wind power data forecasting. *Applied Energy* 377 (2025), 124492.
- [70] Seongjun Yun, Minbyul Jeong, et al. 2019. Graph transformer networks. In *Proceedings of the 2019 NeurIPS*, Vol. 32. Curran Associates, Inc.
- [71] Junhan Yang, Zheng Liu, et al. 2021. Graphformers: GNN-nested transformers for representation learning on textual graph. In *Proceedings of the 2021 NeurIPS*, Vol. 34. Curran Associates, Inc., 28798–28810.
- [72] Yunhao Zhang and Junchi Yan. 2023. Crossformer: Transformer utilizing cross-dimension dependency for multivariate time series forecasting. In *Proceedings of the 2023 ICLR*.
- [73] Azul Garza, Cristian Challu, et al. 2024. TimeGPT-1. arXiv:2310.03589. Retrieved from <https://arxiv.org/abs/2310.03589>
- [74] Tian Zhou, Peisong Niu, et al. 2023. One fits all: Power general time series analysis by pretrained LM. In *Proceedings of the 2023 Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 36. Curran Associates, Inc., 43322–43355.
- [75] Abhimanyu Das, Weihao Kong, et al. 2024. A decoder-only foundation model for time-series forecasting. In *Proceedings of the 2024 International Conference on Machine Learning (ICML)*. ACM.
- [76] Xinli Yu, Zheng Chen, et al. 2023. Temporal data meets LLM – Explainable financial time series forecasting. arXiv:2306.11025. Retrieved from <https://arxiv.org/abs/2306.11025>
- [77] Ming Jin, Shiyu Wang, et al. 2023. Time-LLM: Time series forecasting by reprogramming large language models. arXiv:2310.01728. Retrieved from <https://arxiv.org/abs/2310.01728>
- [78] Che Liu, Zhongwei Wan, et al. 2024. ETP: Learning transferable ECG representations via ECG-text pre-training. In *Proceedings of the 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*. 8230–8234.
- [79] Chenxi Sun, Hongyan Li, et al. 2024. TEST: Text prototype aligned embedding to activate LLM’s ability for time series. arXiv:2308.08241. Retrieved from <https://arxiv.org/abs/2308.08241>
- [80] Jielin Qiu, William Han, et al. 2023. Can brain signals reveal inner alignment with human languages?. In *Proceedings of the Findings of the Association for Computational Linguistics: EMNLP 2023*. ACL, 1789–1804.
- [81] Kaifeng Bi, Lingxi Xie, et al. 2023. Accurate medium-range global weather forecasting with 3D neural networks. *Nature* 619, 7970 (2023), 533–538.
- [82] Kang Chen, Tao Han, et al. 2023. FengWu: Pushing the skillful global medium-range weather forecast beyond 10 days lead. arXiv:2304.02948. Retrieved from <https://arxiv.org/abs/2304.02948>
- [83] Yunkai Zhang, Yawen Zhang, et al. 2023. Insight miner: A time series analysis dataset for cross-domain alignment with natural language. In *Proceedings of the 2023 NeurIPS Workshop AI4Science*. Curran Associates, Inc.
- [84] Alexey Dosovitskiy, Lucas Beyer, et al. 2021. An image is worth 16x16 words: Transformers for image recognition at scale. In *Proceedings of the 2021 International Conference on Learning Representations (ICLR)*.
- [85] Yixuan Su, Tian Lan, et al. 2023. PandaGPT: One model to instruction-follow them all. In *Proceedings of the 1st Workshop on Taming Large Language Models: Controllability in the Era of Interactive Assistants!* ACL, 11–23.
- [86] Ling Cai, Krzysztof Janowicz, et al. 2020. Traffic transformer: Capturing the continuity and periodicity of time series for traffic forecasting. *Transactions in GIS* 24, 3 (2020), 736–755.
- [87] Haoyang Yan, Xiaolei Ma, and Ziyuan Pu. 2021. Learning dynamic and hierarchical traffic spatiotemporal features with transformer. *IEEE Transactions on Intelligent Transportation Systems* 23, 11 (2021), 22386–22399.
- [88] Yuchen Fang, Fang Zhao, et al. 2022. Learning all dynamics: Traffic forecasting via locality-aware spatio-temporal joint transformer. *IEEE Transactions on Intelligent Transportation Systems* 23, 12 (2022), 23433–23446.

- [89] Boyu Huang, Haowen Dou, et al. 2022. Adaptive spatiotemporal transformer graph network for traffic flow forecasting by IoT loop detectors. *IEEE Internet of Things Journal* 10, 2 (2022), 1642–1653.
- [90] Aosong Feng and Leandros Tassioulas. 2022. Adaptive graph spatial-temporal transformer network for traffic forecasting. In *Proceedings of the 2022 ACM CIKM*. ACM, 3933–3937.
- [91] Lei Huang, Feng Mao, et al. 2022. Spatial-temporal convolutional transformer network for multivariate time series forecasting. *Sensors* 22, 3 (2022), 841.
- [92] Zewei Zhao, Xichao Dong, et al. 2024. MDTNet: Multi-scale deformable transformer network with fourier space losses toward fine-scale spatiotemporal precipitation nowcasting. *IEEE Transactions on Geoscience and Remote Sensing* (2024), 1–1.
- [93] Hao Zhang, Yajie Zou, et al. 2022. A temporal fusion transformer for short-term freeway traffic speed multistep prediction. *Neurocomputing* 500 (2022), 329–340.
- [94] Shengnan Guo, Youfang Lin, et al. 2019. Attention based spatial-temporal graph convolutional networks for traffic flow forecasting. In *Proceedings of the 2019 AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 33. AAAI, 922–929.
- [95] Xiaozhuang Song, Ying Wu, and Chenhan Zhang. 2021. TSTNet: A sequence to sequence transformer network for spatial-temporal traffic prediction. In *Proceedings of the 2021 ICANN*. Springer, 343–354.
- [96] Bin Pu, Jiansong Liu, et al. 2024. MVSTT: A multiview spatial-temporal transformer network for traffic-flow forecasting. *IEEE Transactions on Cybernetics* 54, 3 (2024), 1582–1595.
- [97] Changlu Chen, Yanbin Liu, et al. 2022. Bidirectional spatial-temporal adaptive transformer for urban traffic flow forecasting. *IEEE Transactions on Neural Networks and Learning Systems* 34, 10 (2022), 6913–6925.
- [98] Guangyu Huo, Yong Zhang, et al. 2023. Hierarchical spatio-temporal graph convolutional networks and transformer network for traffic flow forecasting. *IEEE Transactions on Intelligent Transportation Systems* 24, 4 (2023), 3855–3867.
- [99] Qianqian Ren, Yang Li, and Yong Liu. 2023. Transformer-enhanced periodic temporal convolution network for long short-term traffic flow forecasting. *Expert Systems with Applications* 227 (2023), 120203.
- [100] Weiguo Zhu, Yongqi Sun, et al. 2023. A correlation information-based spatiotemporal network for traffic flow forecasting. *Neural Computing and Applications* 35, 28 (2023), 21181–21199.
- [101] Wenying Duan, Liu Jiang, et al. 2019. Pre-trained bidirectional temporal representation for crowd flows prediction in regular region. *IEEE Access* 7 (2019), 143855–143865.
- [102] Jiexin Wang, Adam Jatowt, et al. 2023. Bitimebert: Extending pre-trained language representations with bi-temporal information. In *Proceedings of the 2023 ACM SIGIR*. ACM, 812–821.
- [103] Xuhong Wang, Ding Wang, et al. 2023. Building transportation foundation model via generative graph transformer. In *Proceedings of the 2023 IEEE International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 6042–6047.
- [104] Yuxiang Lu, Shalayiding Sirejiding, et al. 2024. Prompt guided transformer for multi-task dense prediction. *IEEE Transactions on Multimedia* 26 (2024), 6375–6385.
- [105] Junhao Zhang, Junjie Tang, et al. 2023. Spatio-temporal pre-training enhanced fast pure transformer network for traffic flow forecasting. In *Proceedings of the 2023 IJCNN*. IEEE, 1–8.
- [106] Xiaoxin Pan, Long Wang, et al. 2022. Short-term wind speed forecasting based on spatial-temporal graph transformer networks. *Energy* 253 (2022), 124095.
- [107] Manzhou Yu, Arif Masrur, and Christopher Blaszczyk-Boxe. 2023. Predicting hourly PM2.5 concentrations in wildfire-prone areas using a SpatioTemporal transformer model. *Science of The Total Environment* 860 (2023), 160446.
- [108] Zikai Song, Junqing Yu, et al. 2022. Transformer tracking with cyclic shifting window attention. In *Proceedings of the 2022 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 8791–8800.
- [109] Zhihan Gao, Xingjian Shi, et al. 2022. Earthformer: Exploring space-time transformers for earth system forecasting. In *Proceedings of the 2022 NeurIPS*, Vol. 35. Curran Associates, Inc., 25390–25403.
- [110] Huijuan Wu et al. 2022. Multistep short-term wind speed forecasting using transformer. *Energy* 261 (2022), 125231.
- [111] Qizhao Jin, Xinbang Zhang, et al. 2024. Preformer: Simple and efficient design for precipitation nowcasting with transformers. *IEEE Geoscience and Remote Sensing Letters* 21 (2024), 1–5.
- [112] Tangjie Wu and Qiang Ling. 2024. Mixformer: Mixture transformer with hierarchical context for spatio-temporal wind speed forecasting. *Energy Conversion and Management* 299 (2024), 117896.
- [113] Fudong Lin, Xu Yuan, et al. 2023. Comprehensive transformer-based model architecture for real-world storm prediction. In *Proceedings of the 2023 ECML PKDD*. Springer, 54–71.
- [114] Wenzhu Li and Shuang Wang. 2022. Federated meta-learning for spatial-temporal prediction. *Neural Computing and Applications* 34, 13 (2022), 10355–10374.
- [115] Tung Nguyen, Johannes Brandstetter, et al. 2023. ClimaX: A foundation model for weather and climate. In *Proceedings of the 2023 International Conference on Machine Learning (ICML)*. ACM, 25904–25938.
- [116] Haoqi Fan, Bo Xiong, et al. 2021. Multiscale vision transformers. In *Proceedings of the 2021 ICCV*. IEEE, 6824–6835.
- [117] Yuxuan Liang, Kun Ouyang, et al. 2022. TrajFormer: Efficient trajectory classification with transformers. In *Proceedings of the 2022 ACM International Conference on Information & Knowledge Management (CIKM)*. ACM, 1229–1237.

- [118] Lina Achaji, Thierno Barry, et al. 2022. PreTR: Spatio-temporal non-autoregressive trajectory prediction transformer. In *Proceedings of the 2022 IEEE International Conference on Intelligent Transportation Systems (ITSC)*. IEEE, 2457–2464.
- [119] Yuqi Chen, Hanyuan Zhang, et al. 2023. RNTrajRec: Road network enhanced trajectory recovery with spatial-temporal transformer. In *Proceedings of the 2023 IEEE International Conference on Data Engineering (ICDE)*. IEEE, 829–842.
- [120] Emre Aksan, Manuel Kaufmann, et al. 2021. A spatio-temporal transformer for 3D human motion prediction. In *Proceedings of the 2021 International Conference on 3D Vision (3DV)*. IEEE, 565–574.
- [121] Hua Yu, Xuanzhe Fan, et al. 2023. Toward realistic 3D human motion prediction with a spatio-temporal cross-transformer approach. *IEEE Transactions on Circuits and Systems for Video Technology* 33, 10 (2023), 5707–5720.
- [122] Trung Le and Eli Shlizerman. 2022. STNDT: Modeling neural population activity with spatiotemporal transformers. In *Proceedings of the 2022 NeurIPS*, Vol. 35. Curran Associates, Inc., 17926–17939.
- [123] Yutian Pang, Xinyu Zhao, et al. 2022. Bayesian spatio-temporal graph transformer network (b-star) for multi-aircraft trajectory prediction. *Knowledge-Based Systems* 249 (2022), 108998.
- [124] Qiheng Mao, Zemin Liu, et al. 2023. HINormer: Representation learning on heterogeneous information networks with graph transformer. In *Proceedings of the 2023 ACM Web Conference (WWW)*. ACM, 599–610.
- [125] Cuiliu Yang and Zhao Pei. 2023. Long-short term spatio-temporal aggregation for trajectory prediction. *IEEE Transactions on Intelligent Transportation Systems* 24, 4 (2023), 4114–4126.
- [126] Liushuai Shi, Le Wang, et al. 2023. Trajectory unified transformer for pedestrian trajectory prediction. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*. IEEE, 9675–9684.
- [127] Xiaolong Tang, Meina Kan, et al. 2024. HPNet: Dynamic trajectory forecasting with historical prediction attention. In *Proceedings of the 2024 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*. IEEE, 15261–15270.
- [128] Maosi Geng, Yong Chen, et al. 2023. Dynamic-learning spatial-temporal transformer network for vehicular trajectory prediction at urban intersections. *Transportation Research Part C: Emerging Technologies* 156 (2023), 104330.
- [129] Yujun Cai, Lin Huang, et al. 2020. Learning progressive joint propagation for human motion prediction. In *Proceedings of the 2020 European Conference on Computer Vision (ECCV)*. Springer, 226–242.
- [130] Qin Li, Yong Wang, and Fanbing Lv. 2024. Semantic correlation attention-based multiorder multiscale feature fusion network for human motion prediction. *IEEE Transactions on Cybernetics* 54, 2 (2024), 825–838.
- [131] Yao Liu, Lina Yao, et al. 2022. Social graph transformer networks for pedestrian trajectory prediction in complex social scenarios. In *Proceedings of the 2022 ACM CIKM*. ACM, 1339–1349.
- [132] Fan Wen, Ming Li, and Ruiyang Wang. 2022. Social transformer: A pedestrian trajectory prediction method based on social feature processing using transformer. In *Proceedings of the 2022 IJCNN*. IEEE, 1–7.
- [133] Pengqin Wang, Meixin Zhu, et al. 2023. BEVGPT: Generative pre-trained large model for autonomous driving prediction, decision-making, and planning. arXiv:2310.10357. Retrieved from <https://arxiv.org/abs/2310.10357>
- [134] Rashid Ali and Hsu-Feng Hsiao. 2023. Enhanced pedestrian trajectory prediction via the cross-modal feature fusion transformer. In *Proceedings of the 2023 VCIP*. IEEE, 1–5.
- [135] Haleh Damirchi, Michael Greenspan, and Ali Etemad. 2023. Context-aware pedestrian trajectory prediction with multimodal transformer. In *Proceedings of the 2023 ICIP*. IEEE, 2535–2539.
- [136] Kimberly Amador, Anthony Winder, et al. 2022. Hybrid spatio-temporal transformer network for predicting ischemic stroke lesion outcomes from 4D CT perfusion imaging. In *Proceedings of the 2022 MICCAI*. Springer, 644–654.
- [137] Nahian Siddique, Sidike Paheding, et al. 2021. U-net and its variants for medical image segmentation: A review of theory and applications. *IEEE Access* 9 (2021), 82031–82057.
- [138] Shilin Sun, Yuekai Liu, et al. 2023. Short-term multi-step wind power forecasting based on spatio-temporal correlations and transformer neural networks. *Energy Conversion and Management* 283 (2023), 116916.
- [139] Zihan Chen, Lei Nico Zheng, et al. 2023. ChatGPT informed graph neural network for stock movement prediction. arXiv:2306.03763. Retrieved from <https://arxiv.org/abs/2306.03763>
- [140] Jin Fan, Wenchao Weng, et al. 2024. RGDAN: A random graph diffusion attention network for traffic prediction. *Neural Networks* 172 (2024), 106093.
- [141] Yanfeng Sun, Xiangheng Jiang, et al. 2022. Dual dynamic spatial-temporal graph convolution network for traffic prediction. *IEEE Transactions on Intelligent Transportation Systems* 23, 12 (2022), 23680–23693.
- [142] Guojian Zou, Ziliang Lai, et al. 2024. MT-STNet: A novel multi-task spatiotemporal network for highway traffic flow prediction. *IEEE Transactions on Intelligent Transportation Systems* 25, 7 (2024).
- [143] Junbo Zhang, Yu Zheng, and Dekang Qi. 2017. Deep spatio-temporal residual networks for citywide crowd flows prediction. In *Proceedings of the 2017 AAAI Conference on Artificial Intelligence (AAAI)*, Vol. 31. AAAI.
- [144] Jian Liang, Jinjun Tang, et al. 2023. On region-level travel demand forecasting using multi-task adaptive graph attention network. *Information Sciences* 622 (2023), 161–177.
- [145] Xu Liu, Yutong Xia, et al. 2024. Largest: A benchmark dataset for large-scale traffic forecasting. In *Proceedings of the 2024 Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 36. Curran Associates, Inc., 75354–75371.

- [146] Joaquín Muñoz-Sabater, Emanuel Dutra, et al. 2021. ERA5-Land: A state-of-the-art global reanalysis dataset for land applications. *Earth System Science Data* 13, 9 (2021), 4349–4383.
- [147] Andrey Malinin, Neil Band, et al. 2021. Shifts: A dataset of real distributional shift across multiple large-scale tasks. In *Proceedings of the 2021 Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 1. Curran Associates, Inc.
- [148] K Krishna Rani Samal, Korra Sathya Babu, et al. 2021. Multi-directional temporal convolutional artificial neural network for PM_{2.5} forecasting with missing values: A deep learning approach. *Urban Climate* 36 (2021), 100800.
- [149] Matthew W. Jones, John T. Abatzoglou, et al. 2022. Global and regional trends and drivers of fire under climate change. *Reviews of Geophysics* 60, 3 (2022), e2020RG000726.
- [150] Stefano Pellegrini, Andreas Ess, and Luc Van Gool. 2010. Improving data association by joint modeling of pedestrian trajectories and groupings. In *Proceedings of the 2010 ECCV*. Springer, 452–465.
- [151] Amir Sadeghian, Vineet Kosaraju, et al. 2019. Sophie: An attentive GAN for predicting paths compliant to social and physical constraints. In *Proceedings of the 2019 CVPR*. IEEE, 1349–1358.
- [152] Scott Ettinger, Shuyang Cheng, et al. 2021. Large scale interactive motion forecasting for autonomous driving: The waymo open motion dataset. In *Proceedings of the 2021 ICCV*. IEEE, 9710–9719.
- [153] Robert Krajewski et al. 2018. The highD dataset: A drone dataset of naturalistic vehicle trajectories on German highways for validation of highly automated driving systems. In *Proceedings of the 2018 ITSC*. IEEE, 2118–2125.
- [154] Tianrui Guan, Jun Wang, et al. 2022. M3DETR: Multi-representation, multi-scale, mutual-relation 3D object detection with transformers. In *Proceedings of the 2022 WACV*. IEEE, 772–782.
- [155] Chunliang Li, Wencheng Han, et al. 2024. RepVF: A unified vector fields representation for multi-task 3D perception. In *Proceedings of the 2018 European Conference on Computer Vision (ECCV)*. Springer.
- [156] Hao-Shu Fang, Jiefeng Li, et al. 2023. Alphapose: Whole-body regional multi-person pose estimation and tracking in real-time. *IEEE Transactions on Pattern Analysis and Machine Intelligence* 45, 6 (2023), 7157–7173.
- [157] Yufei Zhang, Jeffrey O. Kephart, et al. 2024. PhysPT: Physics-aware pretrained transformer for estimating human dynamics from monocular videos. In *Proceedings of the 2024 CVPR*. IEEE, 2305–2317.
- [158] Zhiyong Cui, Ruimin Ke, et al. 2020. Stacked bidirectional and unidirectional LSTM recurrent neural network for forecasting network-wide traffic state with missing values. *Transportation Research Part C: Emerging Technologies* 118 (2020), 102674.
- [159] Weida Zhong, Qiuling Suo, et al. 2021. Heterogeneous spatio-temporal graph convolution network for traffic forecasting with missing values. In *Proceedings of the 2021 ICDCS*. IEEE, 707–717.
- [160] Jiawei Jiang, Chengkai Han, et al. 2023. Pdfformer: Propagation delay-aware dynamic long-range transformer for traffic flow prediction. In *Proceedings of the 2023 AAAI*, Vol. 37. AAAI, 4365–4373.
- [161] Yixiao Li, Yifan Yu, et al. 2023. LoSparse: Structured compression of large language models based on low-rank and sparse approximation. In *Proceedings of the 2023 ICML*, Vol. 202. ACM, 20336–20350.
- [162] Dan Friedman, Alexander Wettig, and Danqi Chen. 2023. Learning transformer programs. In *Proceedings of the 2023 Advances in Neural Information Processing Systems (NeurIPS)*, Vol. 36. Curran Associates, Inc., 49044–49067.
- [163] Dimitris Bertsimas, Agni Orfanoudaki, and Holly Wiberg. 2021. Interpretable clustering: An optimization approach. *Machine Learning* 110, 1 (2021), 89–138.
- [164] Kun Wang, Hao Wu, et al. 2024. NuwaDynamics: Discovering and updating in causal spatio-temporal modeling. In *Proceedings of the 2024 International Conference on Learning Representations (ICLR)*.

Received 21 September 2024; revised 4 June 2025; accepted 3 September 2025